

Neural Representation Learning for Speech and Audio Signals

PhD Colloquium
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Purvi Agrawal

Advisor: Dr. Sriram Ganapathy

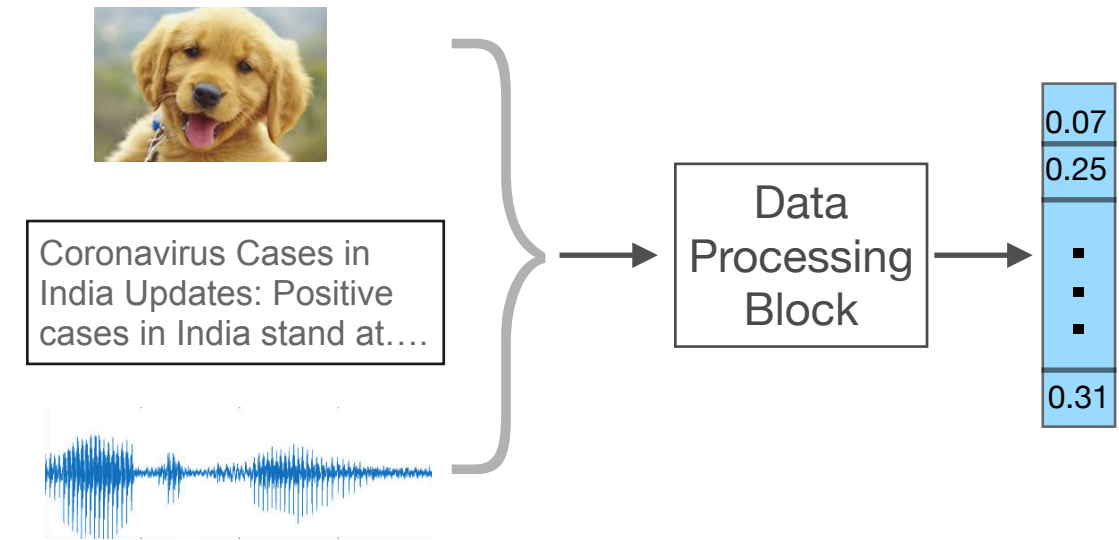
Learning and Extraction of Acoustic Patterns (LEAP) Lab,
Electrical Engg., Indian Institute of Science, Bangalore.



Representation Learning

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- **Representation** - converting raw data to numerical form
- Goal of representation - Succinctly characterizing data (while being robust)
- Traditional approaches
 - Knowledge driven & data-driven learning
- **Representation learning** - learning meaningful representations from raw data
- Motivation - the learning algorithms to be less dependent on feature engineering
- ML algorithms have preprocessing pipelines and data transformations

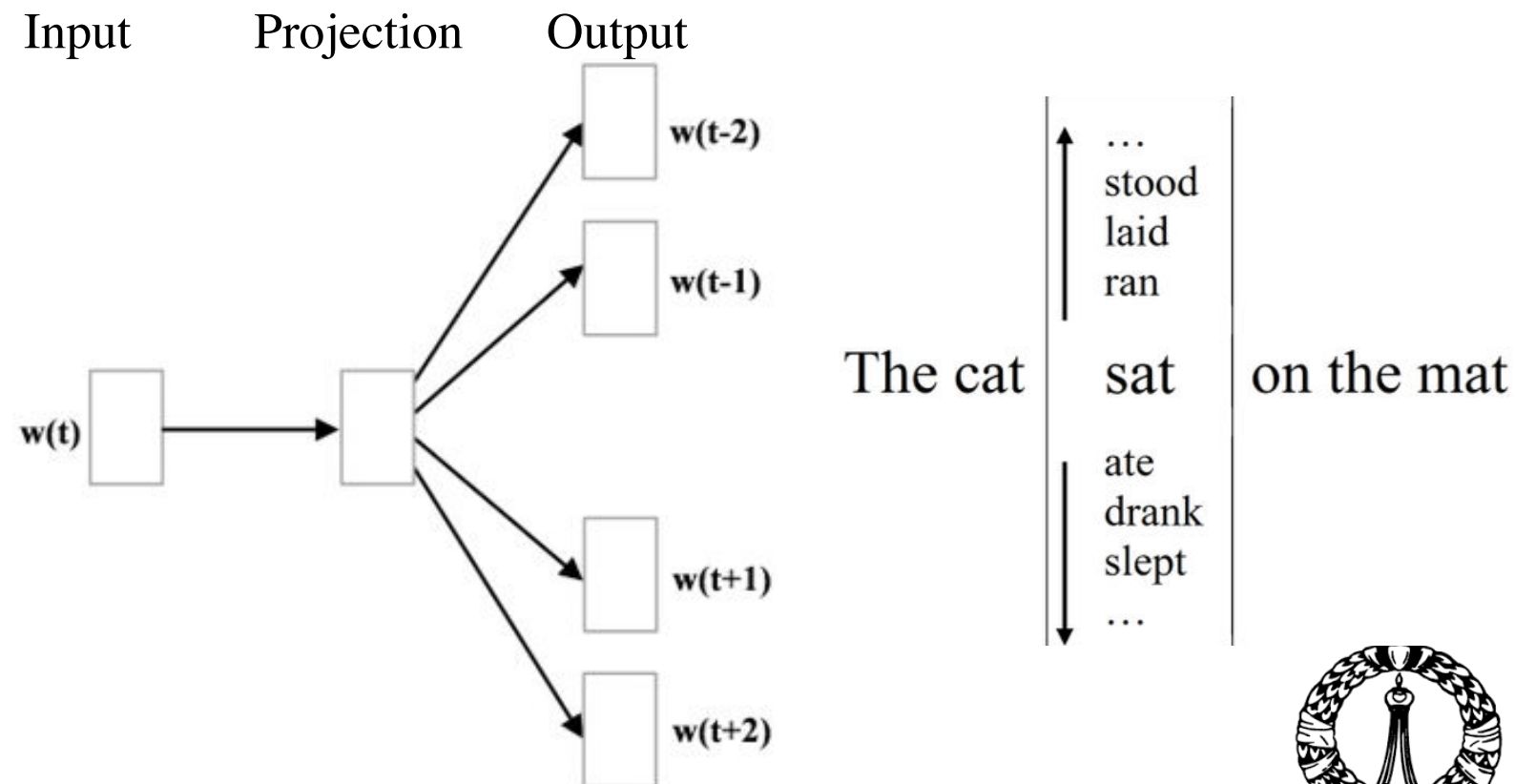


Representation Learning

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- Representation learning using traditional methods:
 - unsupervised manner like principal component analysis (PCA), t-stochastic neighbourhood embeddings (t-SNE), etc
 - supervised manner like linear discriminant analysis (LDA)
- With neural (networks) approaches - **Neural Representation Learning**

- An example in text domain:
Word2vec
 - Converts words to 1-D vectors
 - Creates embeddings for text data to predict next word



Deep Representation Learning

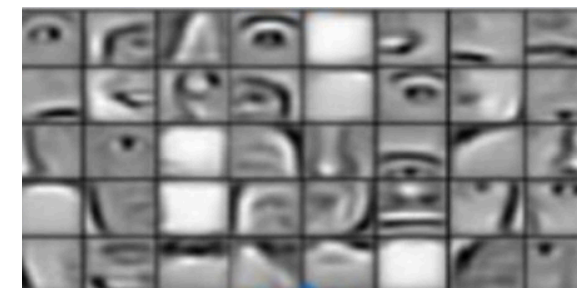
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- Deep representation learning
 - Attempts to learn multiple levels of representation of increasing complexity/abstraction
 - Example: face ID recognition

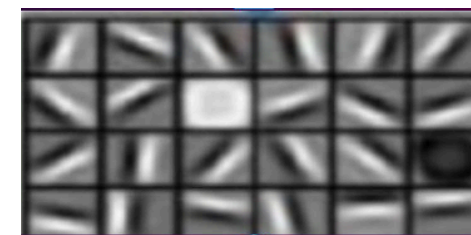


Layer-3
Objects

Parts combine to form objects



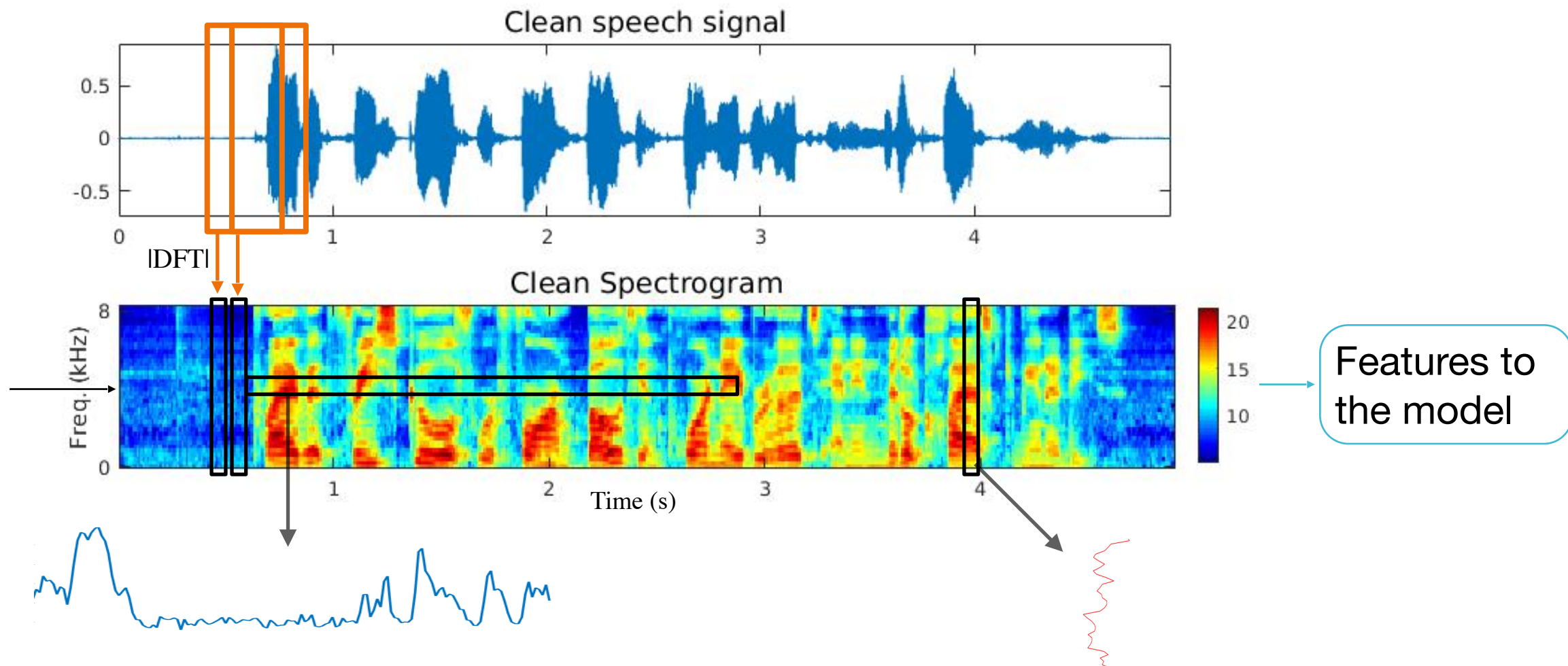
Layer-2
Parts



Layer-1
Edges

Visual pixels

Speech Representation



Temporal modulations (Rate): variation of short-term energy along time, in different frequency subbands

- < 10 Hz capture the syllable rate
- > 10 Hz capture onset and offset transitions

Spectral modulations (Scale): variation of short-term energy along frequency axis, in different time frames

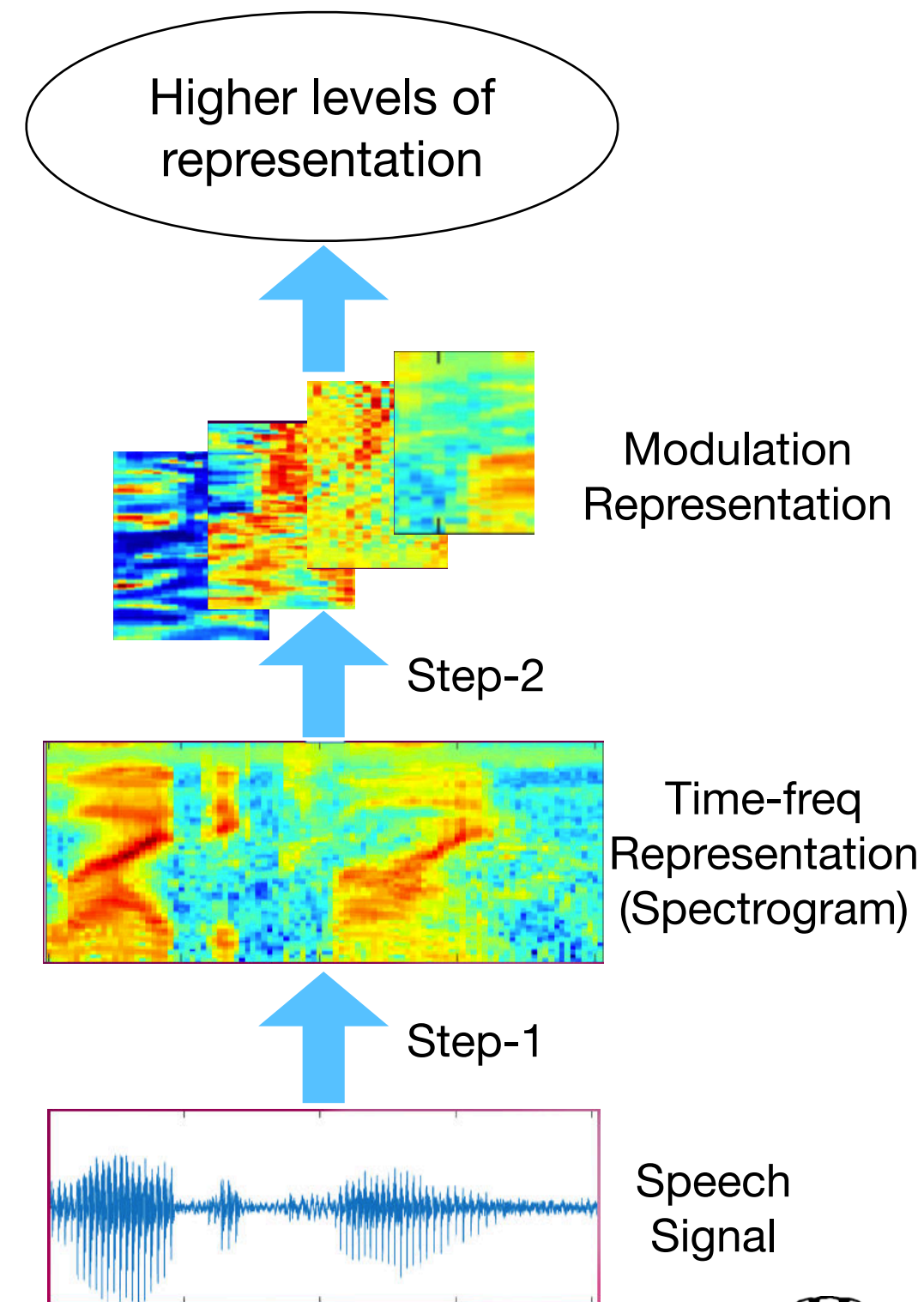
- < 1 cycles/kHz capture the spectral profile
- > 1 cycles/kHz capture fine harmonics details

Modulation filtering - filtering the spectrogram using rate/scale filters to obtain modulation representations

Deep Representation Learning For Speech

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- Step-1: Time-frequency representation
 - Convert the raw waveform into time-frequency representation (short-time spectral energies - spectrogram)
 - Learns acoustic filterbank
- Step 2: Modulation representation
 - Filtering spectrogram using 1-D/2-D modulation filterbank
- Further steps attempts to learn higher levels of complex representation
- Learning framework can be supervised, unsupervised, semi-supervised, transfer learning.



Acoustic filterbank approaches

- Knowledge driven approaches like mel filterbank and gammatone filterbank [Davis & Mermelstein, 1980, Patterson et. al, 1987]
 - Psychoacoustics - mimics early part of human hearing
- Supervised learning
 - Learn kernels of convolution layer from raw waveform [Sainath et. al, 2013, Tuske et. al, 2014]
 - Learn filterbank using power spectra in DNN [Seki et. al., 2017]
 - Interpretable filterbank using SincNet [Ravenelli et. al., 2018]
- Unsupervised learning
 - Using convolutional RBM [Sailor and Patil, 2016]
 - WaveNet auto encoders [Chorowski et. al., 2019]

Modulation filterbank approaches

Perception based experiments on **modulation filtering** range :

- Temporal modulations within 1 – 16 Hz, spectral modulations within 0 – 4 cycles/kHz, are crucial for speech intelligibility [Drullman et. al, 1994].

Knowledge driven modulation filtering:

- RASTA filtering [Hermansky et. al, 1994], Gabor filtering [Kleinschmidt et. al, 2003; Mesgarani et. al., 2004; Domont et. al, 2008]

Unsupervised learning:

- Convolutional deep belief networks for audio classification and speech analysis [Lee et. al., 2009]
- Learning spectro-temporal receptive fields (STRFs) using convolutional RBMs [Sailor et. al., 2016]

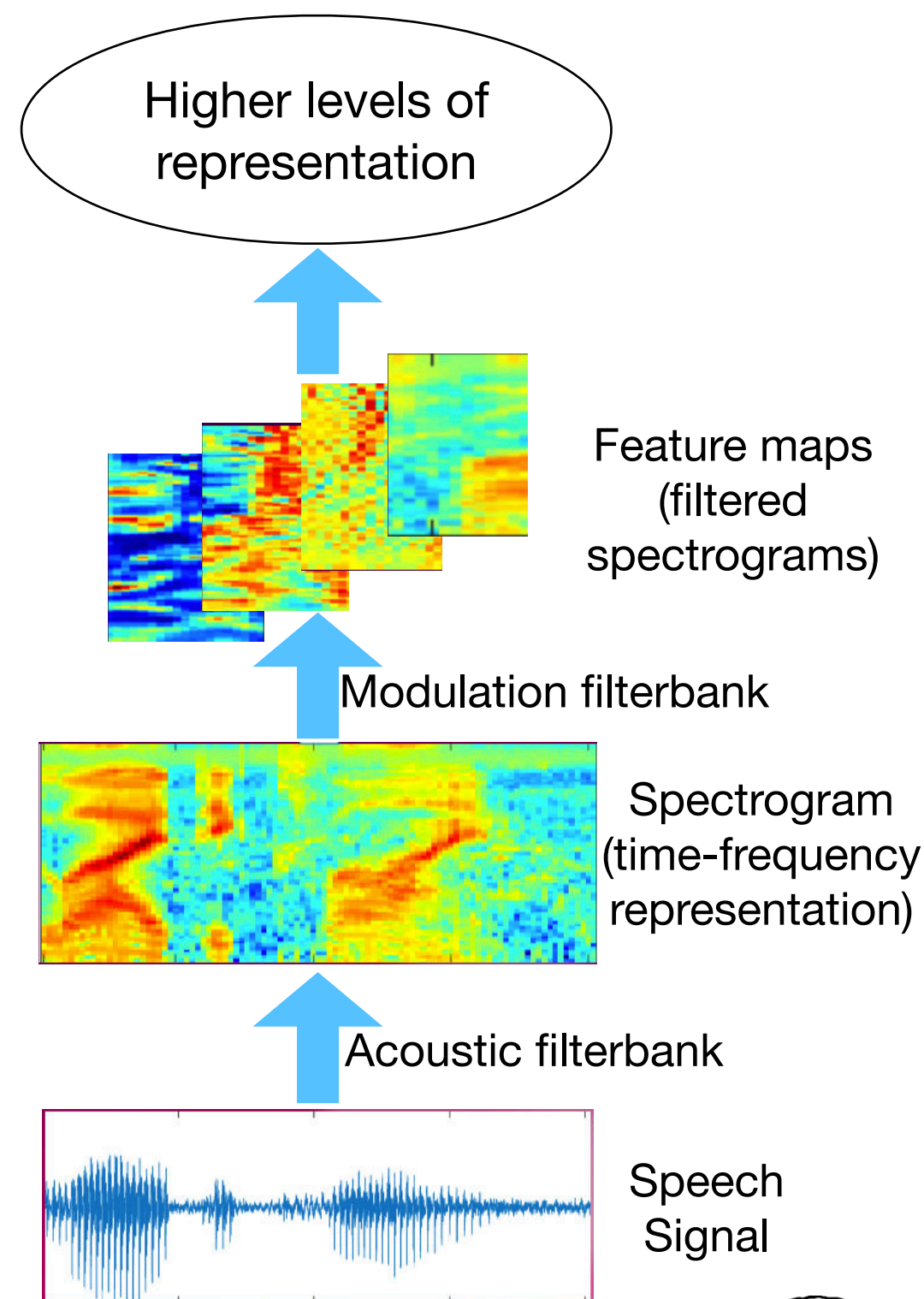
Supervised learning:

- Convolutional neural networks (CNNs) [LeCun and Bengio, 1995]

Some Questions

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- Acoustic filterbank - can we learn from data instead of using handcrafted mel filterbank ?
 - Yes. How do we interpret them?
 - Do we need to weigh all sub-bands in time-frequency representation equally?
- Modulation filterbank - can we learn from data?
 - Yes. How do we interpret them?
 - Can we learn filters with less redundancy from the data?
 - Do we need to weigh all modulation filters equally?
- Interpretability
 - Can we interpret the network parameters?
 - Can we analyse what characteristics of speech/audio signals helped in the task?



This Thesis

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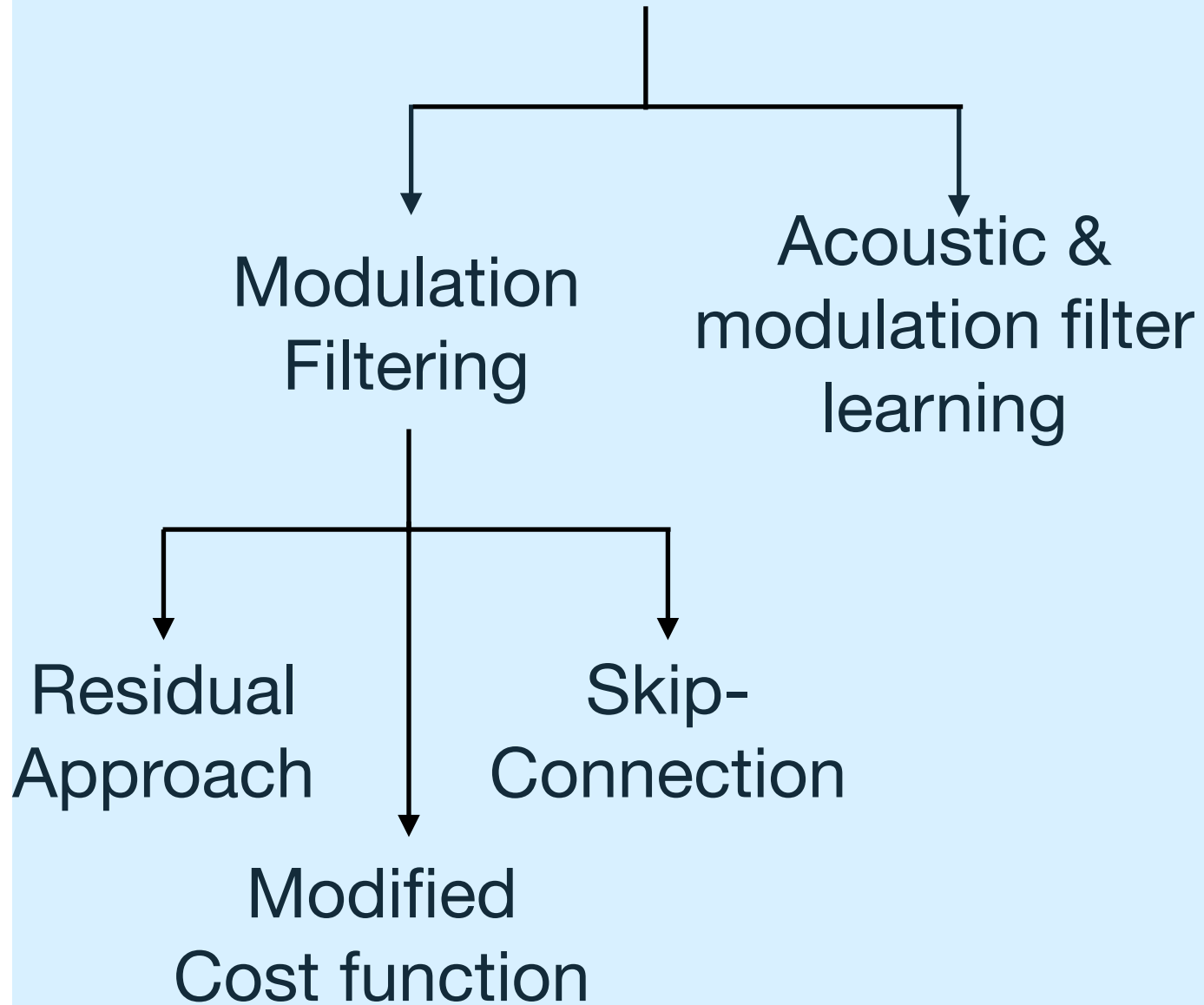
- Neural representation learning of the speech data with focus on initial 2 stages
- **Unsupervised representation learning**
 - Stage-1: Acoustic representation from raw speech waveform
 - Stage-2: Modulation filtered representation from spectrogram features
 - Learn irredundant modulation filters from the data
- **Supervised representation learning**
 - Interpretable representation learning
 - Stage-1: Acoustic representation from raw speech waveform
 - Stage-2: Modulation filtered representation from spectrogram features
 - Relevance weighting scheme to analyse the acoustic/modulation characteristics
 - Extending the approach to audio signals

“Instead of sitting around trying to design new feature detectors, you design new architectures that learn the features” - Larry Davis, Prof UMD & Senior Principal Scientist, Amazon

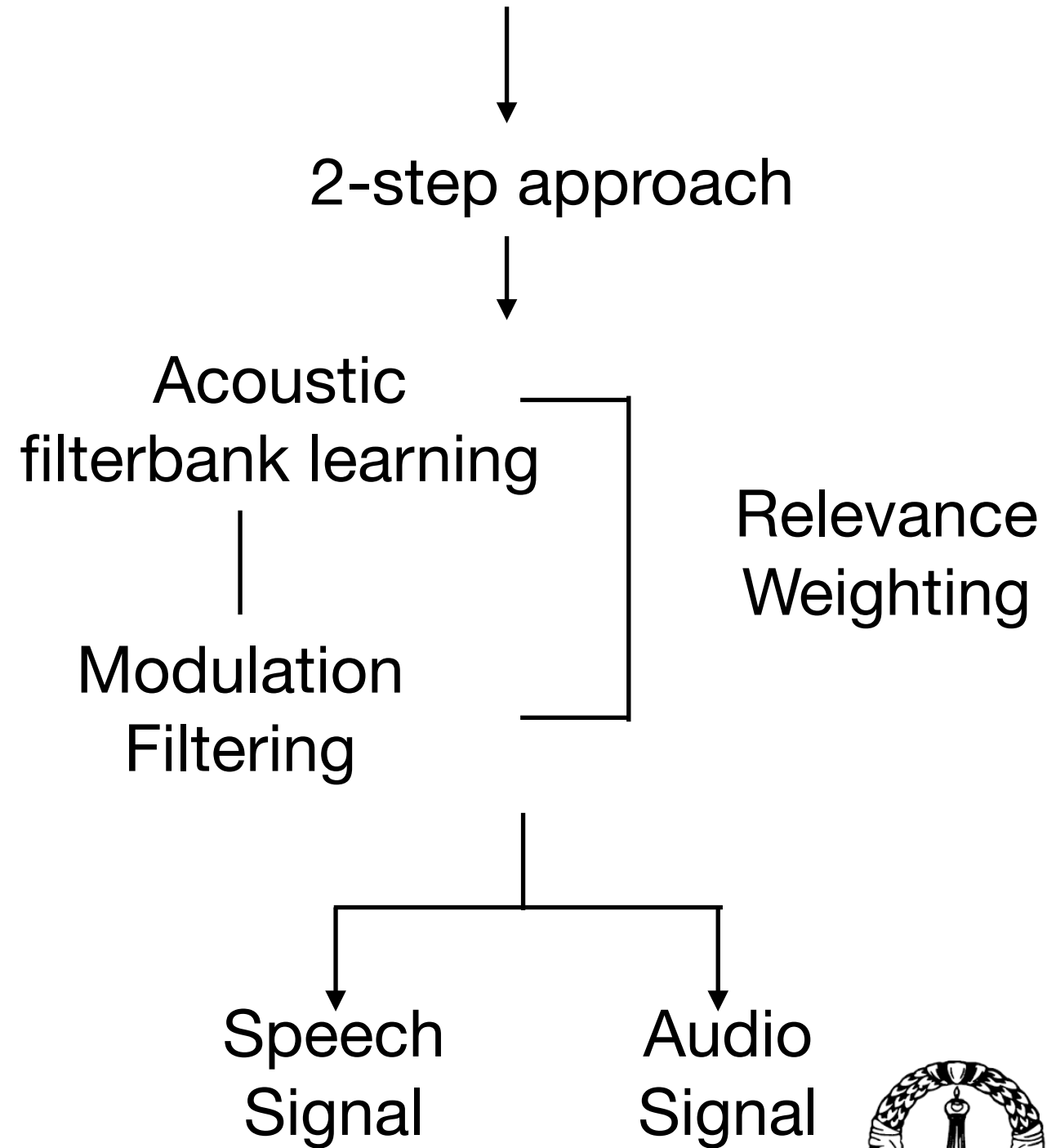
Thesis Map

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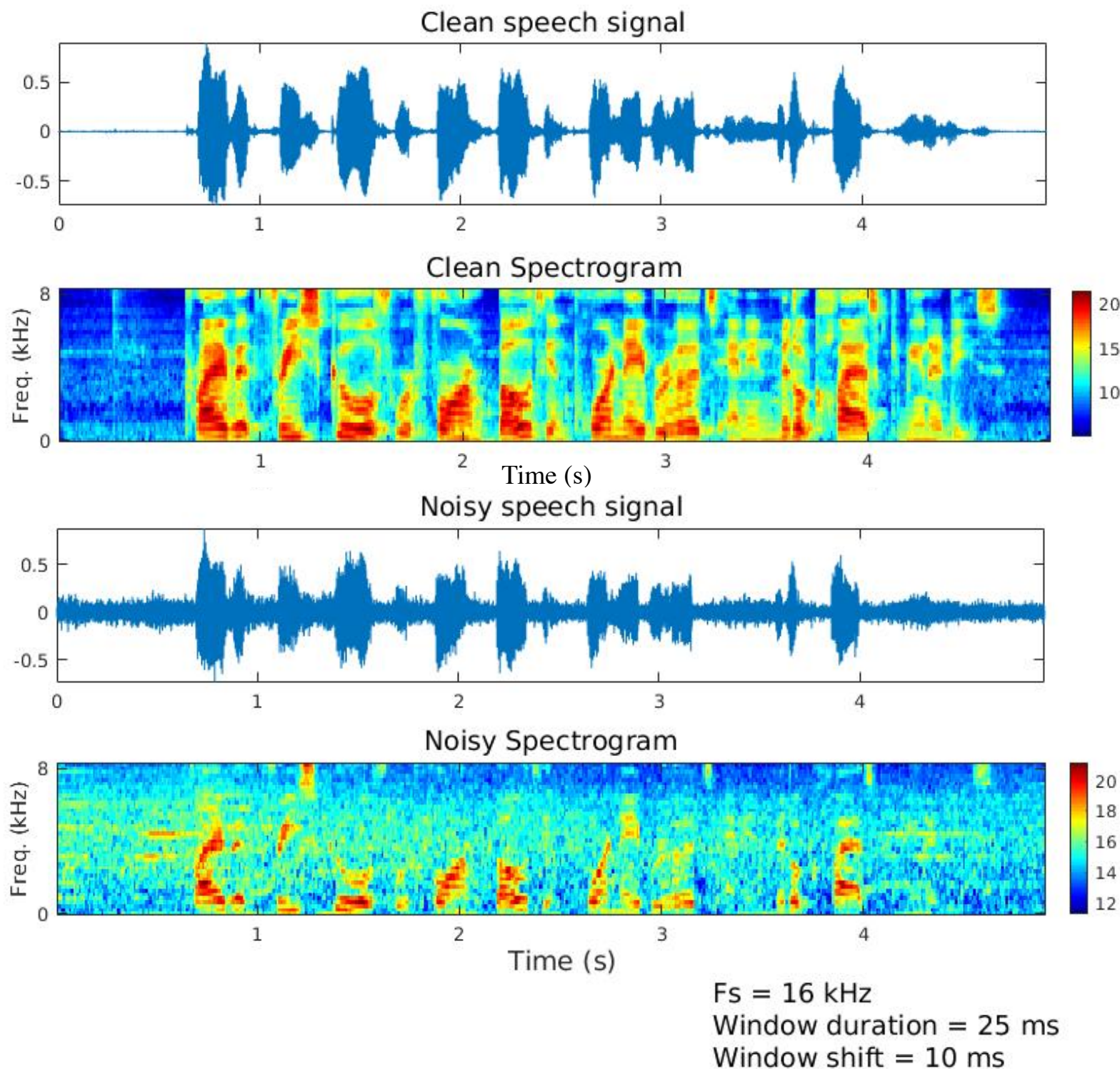
Part-I Unsupervised Learning (Speech)



Part-II Supervised Learning



Effect of distortion



uses Mel filterbank

Features to the model

Distortions in recorded signal:

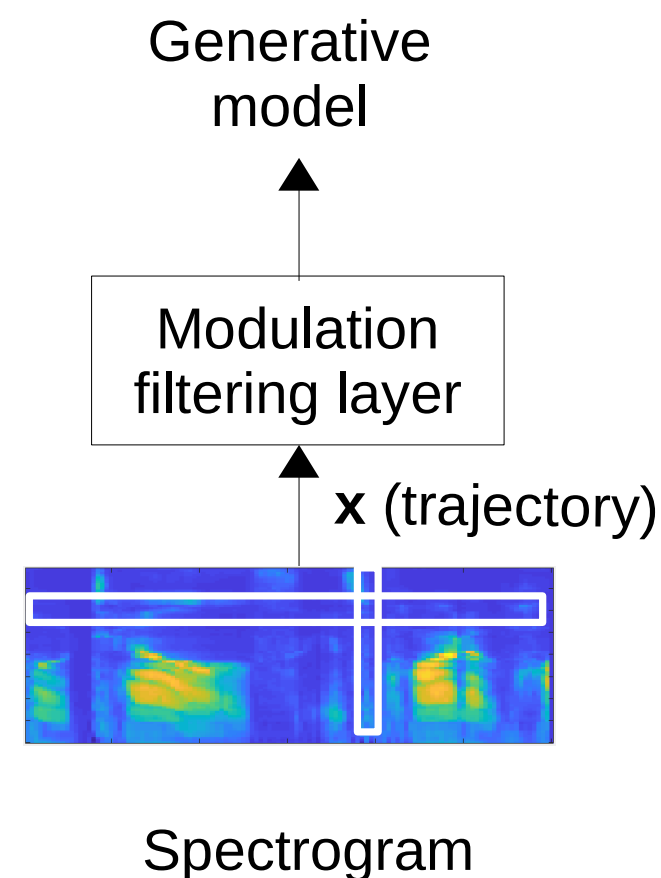
- Signals from speech and different noises received at a microphone: restaurant, train, airport, etc. - Additive noise
- Speech signal and its reflections from surroundings (eg. walls) received at the microphone - Reverberation

Distortion robust representations?

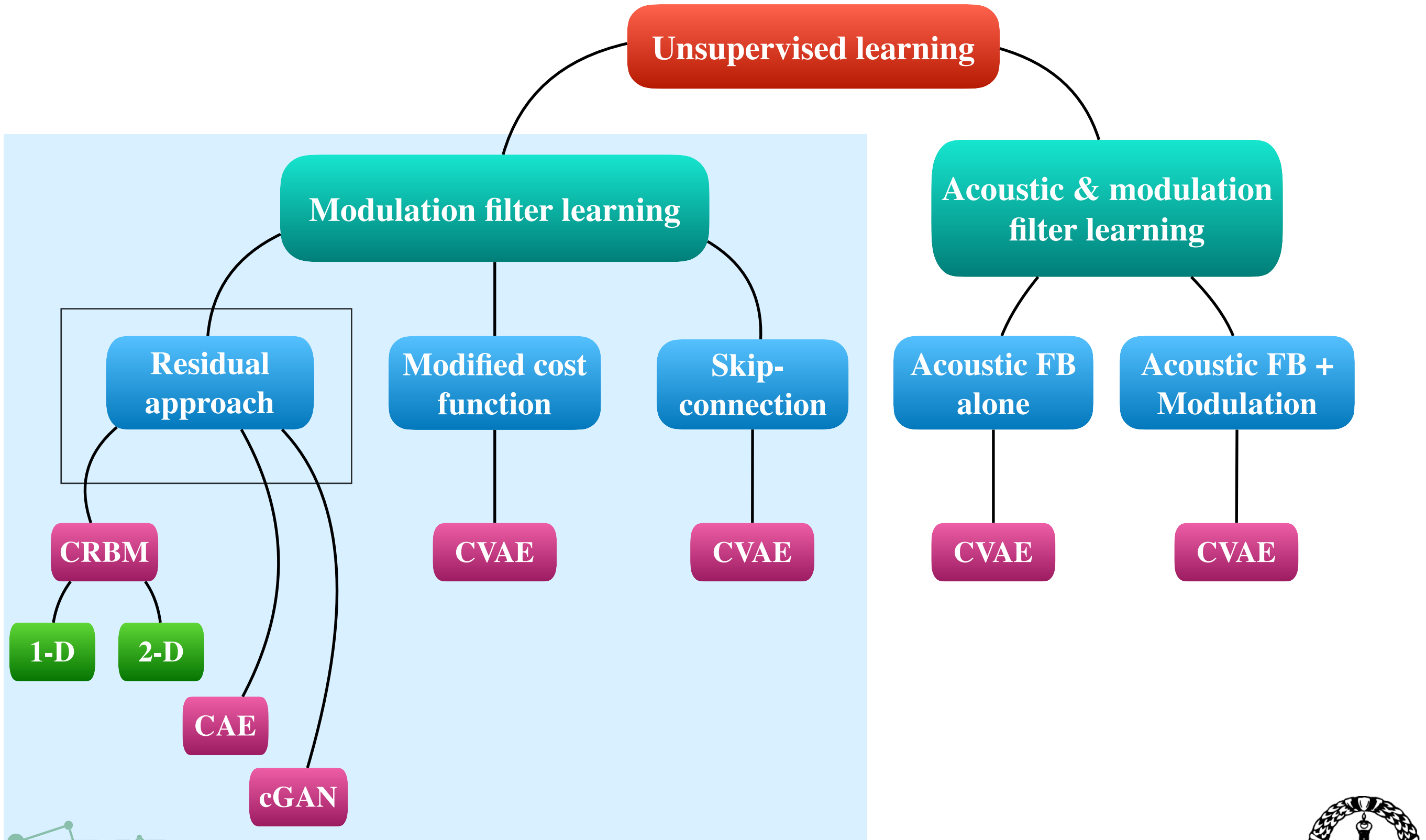
Modulation Filtering

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- Modulation filtering - Involves filtering the spectrogram to retain relevant speech information
- 1-D filters - Filters temporal and spectral 1-D trajectories
- 2-D filters - Filters spectrogram 2-D patch
- With multiple filters - we obtain multiple representations
 - Learn irredundant representations
- Generative models
 - Restricted Boltzmann machine (RBM)
 - Autoencoder (AE), variational AE (VAE)
 - Generative adversarial network (GAN)



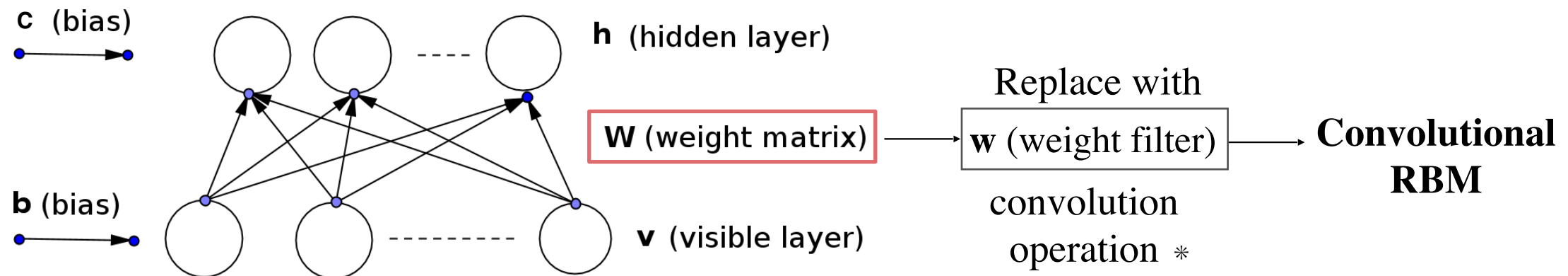
Part I: Unsupervised Learning (Speech)



Restricted Boltzmann Machine (RBM)

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- Convolutional RBM (CRBM)



- Energy based model - tries to minimise pre-defined energy function
- The energy function of the Gaussian CRBM is given as:

$$E(\mathbf{v}, \mathbf{h}, \theta) = \sum_i \frac{1}{2} \mathbf{v}_i^2 - \sum_{i,j} \mathbf{v}_i \mathbf{W}_{ij} \mathbf{h}_j - \sum_j \mathbf{c}_j \mathbf{h}_j$$

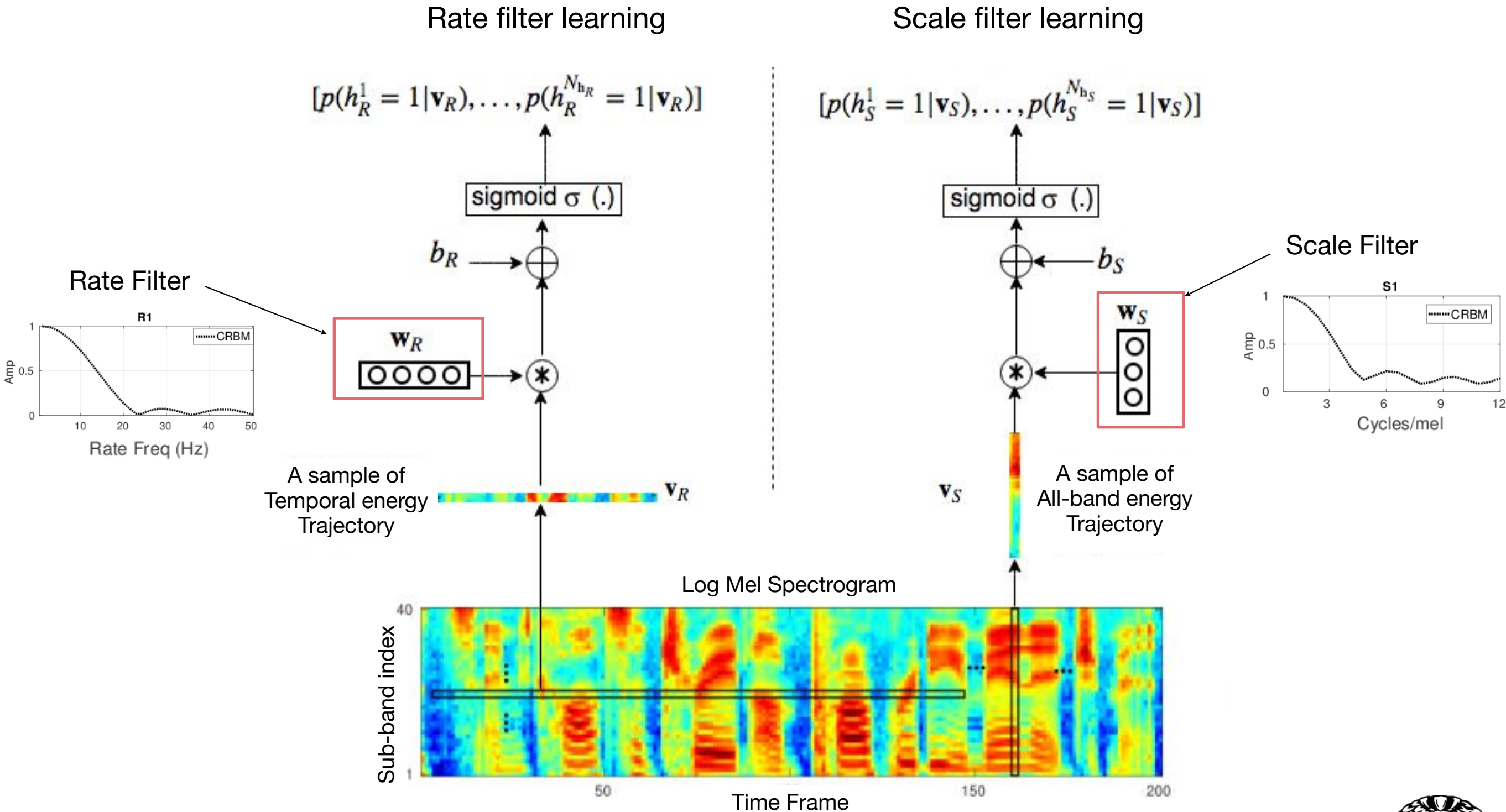
$\downarrow$
 $(\mathbf{v} * \mathbf{w})_i$

- The probability density is given as:

$$P(\mathbf{v}, \theta) = \frac{1}{Z(\theta)} \sum_{\mathbf{h}} e^{-E(\mathbf{v}, \mathbf{h}, \theta)} \quad \mathbf{w}^* = \arg \max_{\mathbf{w}} \sum_{\mathbf{h}} P(\mathbf{v}, \mathbf{h}) \sim \text{Boltzmann Distribution}$$

Learning 1-D Filter Using CRBM

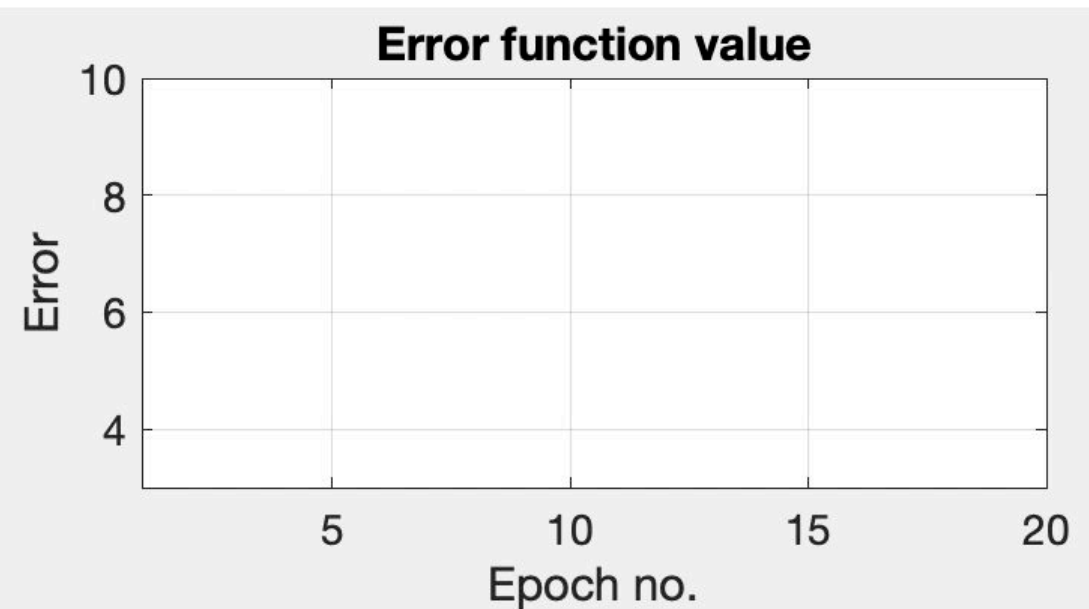
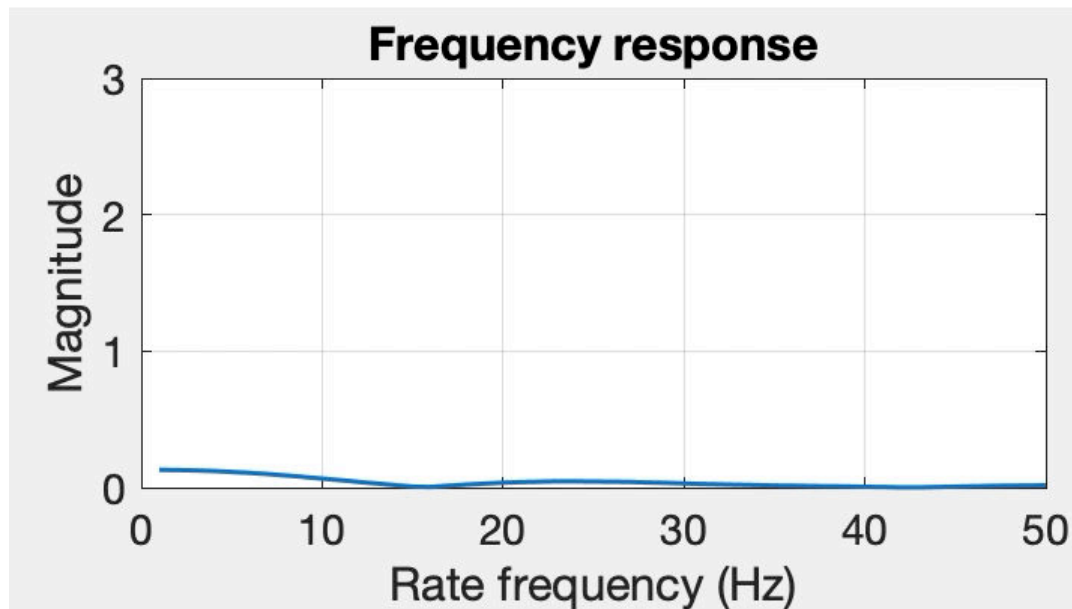
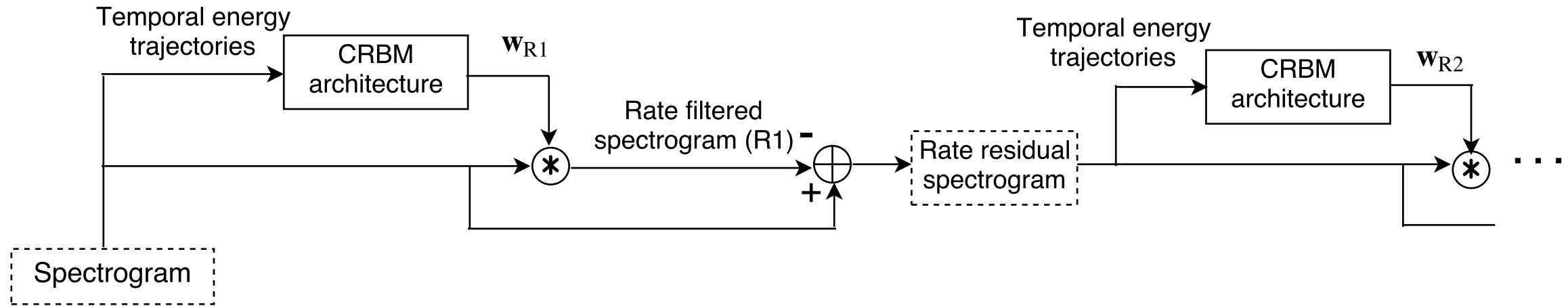
16



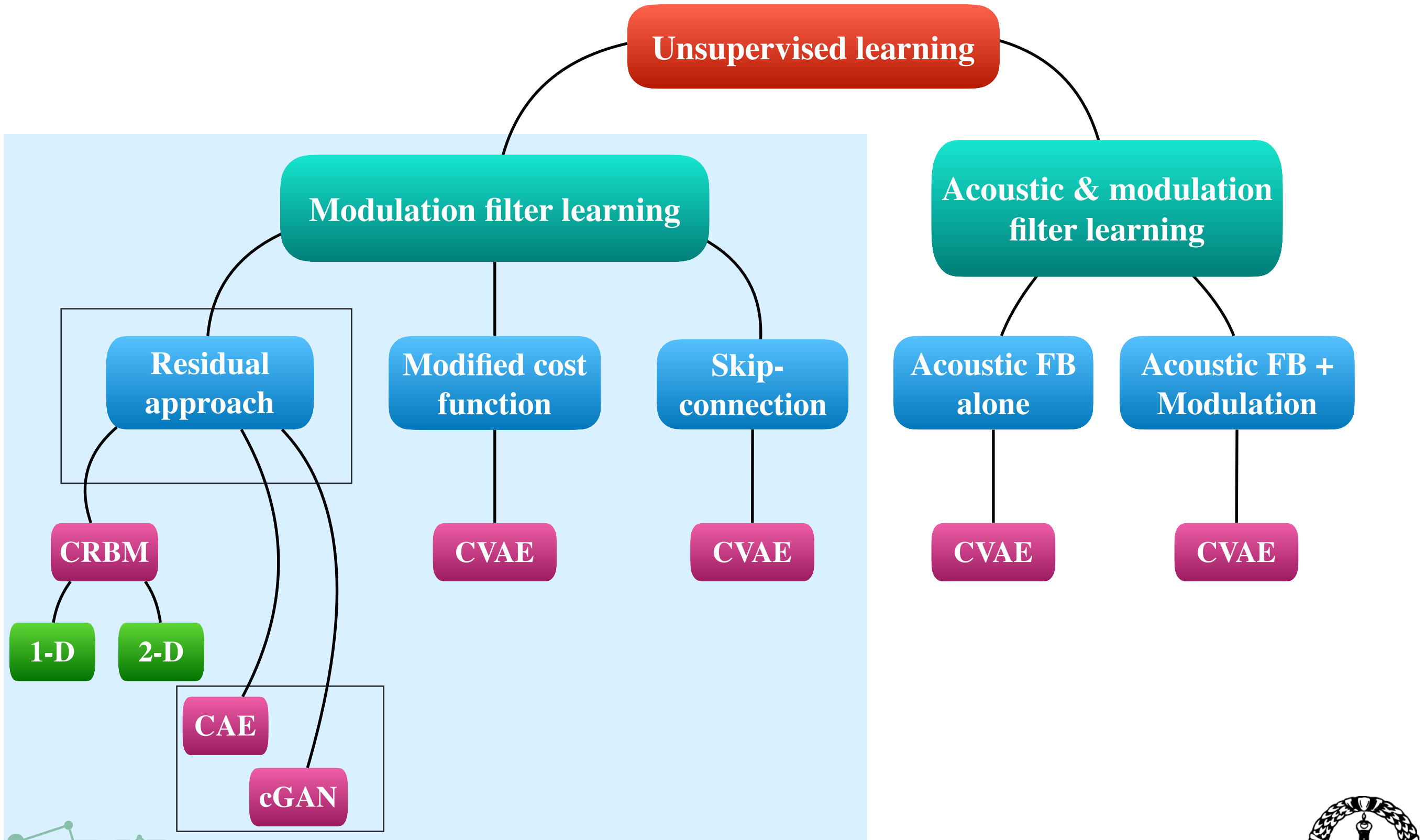
Learning Multiple Filters

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- Residual approach

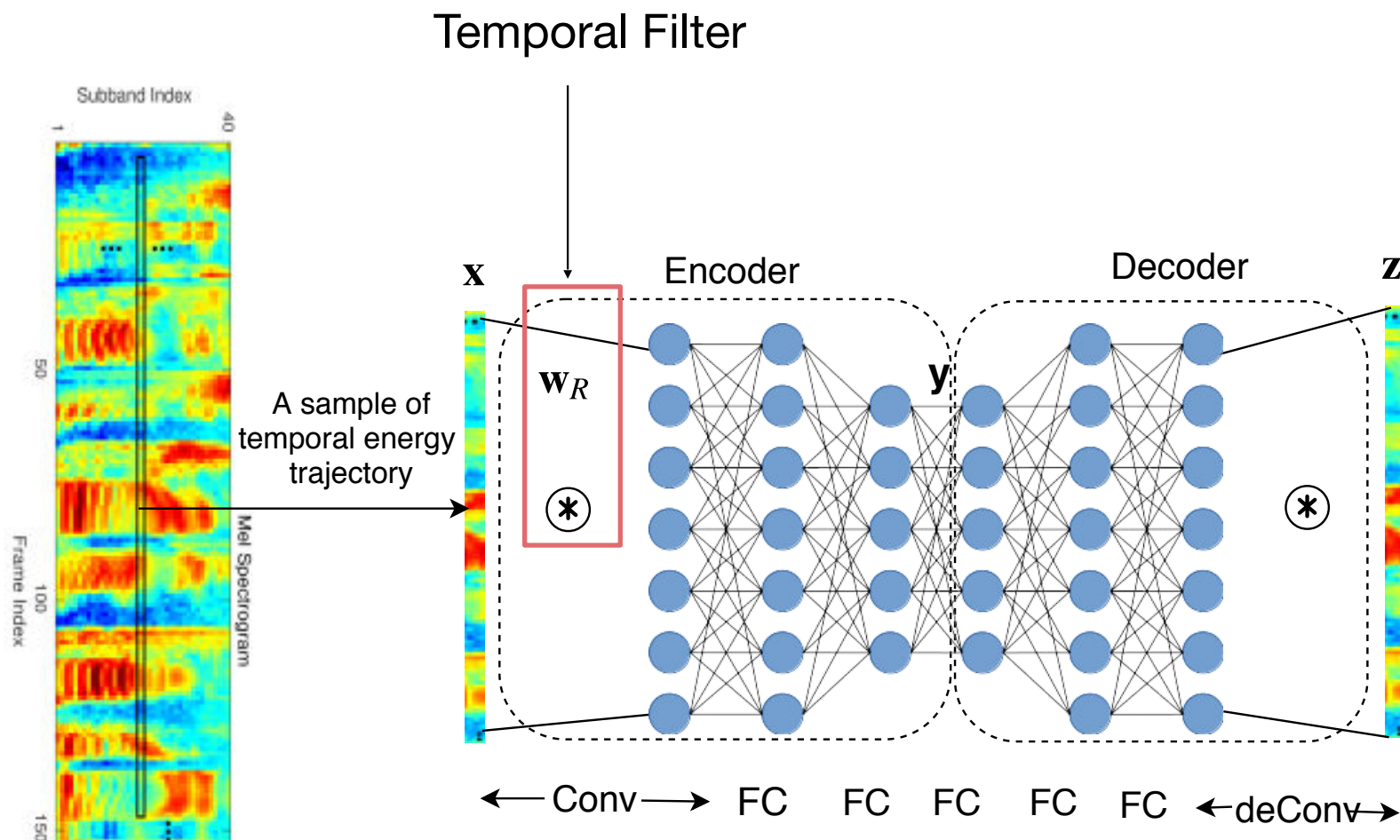


Part I: Unsupervised Learning (Speech)



Convolutional Autoencoder (CAE)

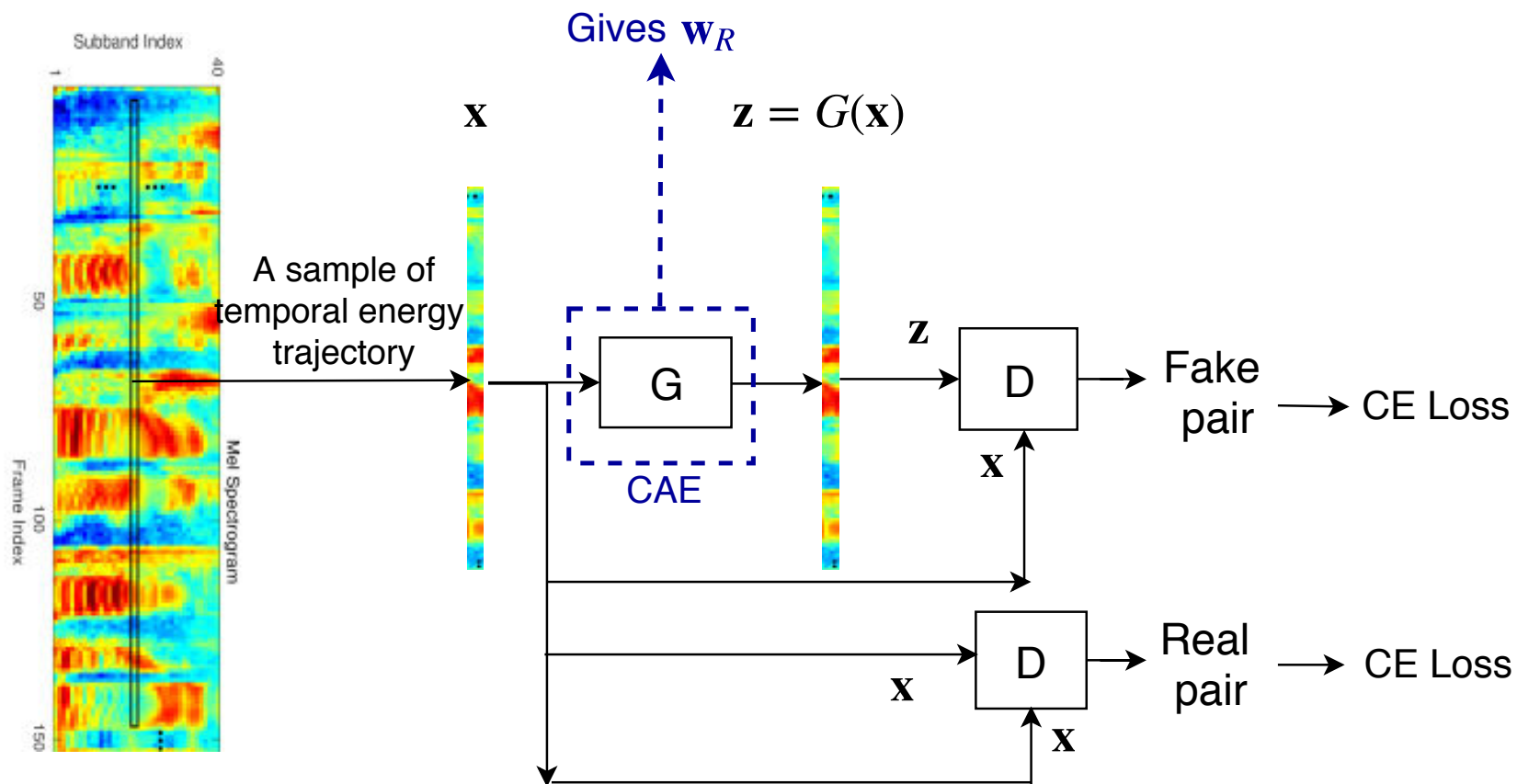
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- Neural network that can convert high-dimensional data to low-dimensional codes using encoder [Hinton et. al, 2006]
- Use a decoder block to reconstruct the data back
- Trained with the objective of minimising $L(\mathbf{x}, \mathbf{z}) \approx ||\mathbf{x} - \mathbf{z}||^2$
- Hence $\mathbf{w}_R^* = \arg \min_{\mathbf{w}_R} L(\mathbf{x}, \mathbf{z})$
- Used for 1-D filter learning

Generative Adversarial Network (GAN)

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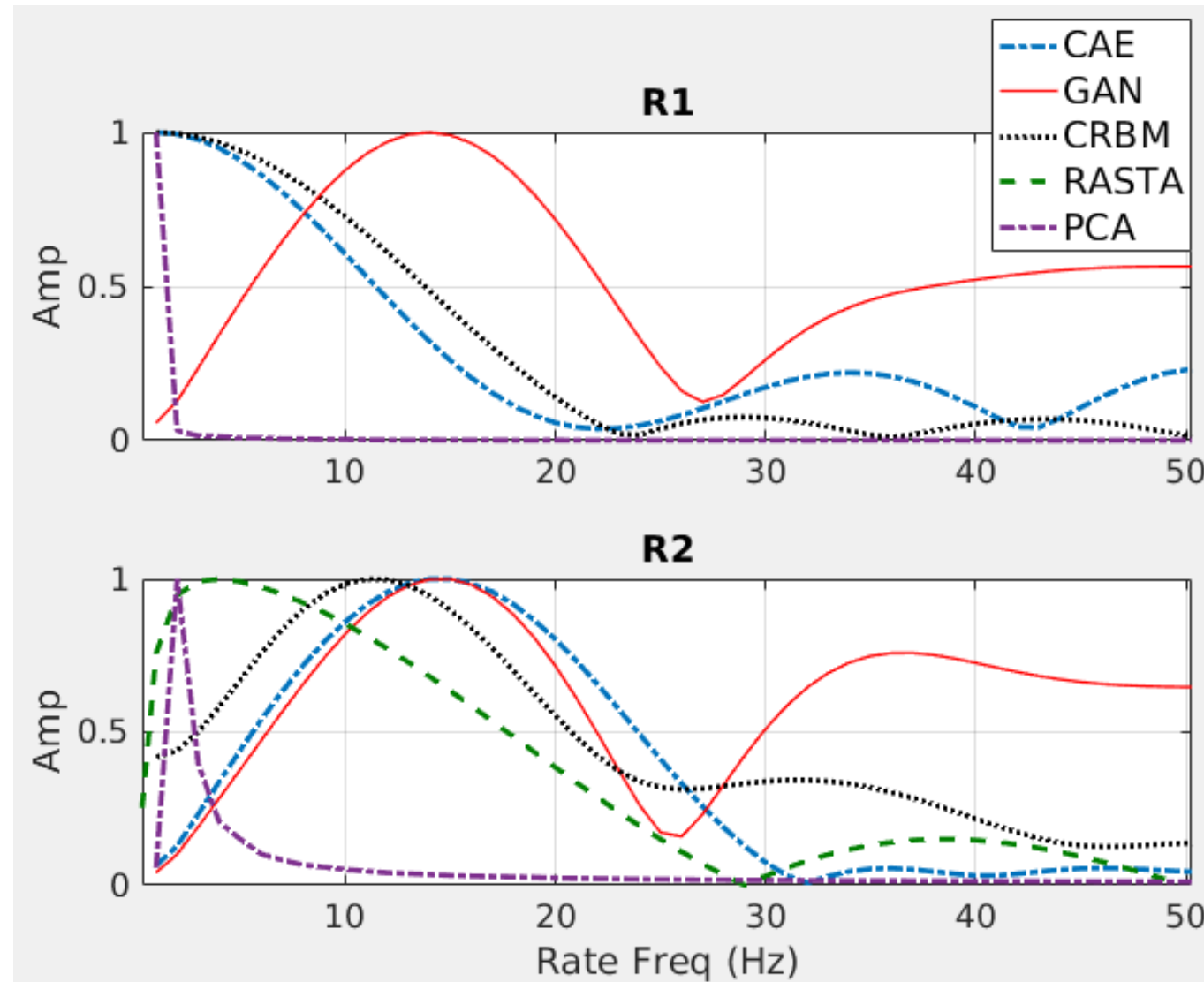
- Model that learn to produce realistic samples of input data via an adversarial learning
- Consists of two blocks : Generator CAE and Discriminator D
- We use conditional GAN (cGAN)
- D estimates probability that the input pair is real pair rather than fake
- CAE tries to maximize the probability of D making a mistake
- The cGAN is trained with the objective of:

$$L_{cGAN}(D, G) = E_{\mathbf{x}}[\log D(\mathbf{x}, \mathbf{x})] + E_{\mathbf{x}}[\log(1 - D(\mathbf{x}, G(\mathbf{x})))] ,$$

$$G^* = \arg \min_G \max_D L_{cGAN}(D, G).$$

Learnt 1-D Filters

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Residual
approach

Frequency response of the learnt temporal modulation (rate) filters

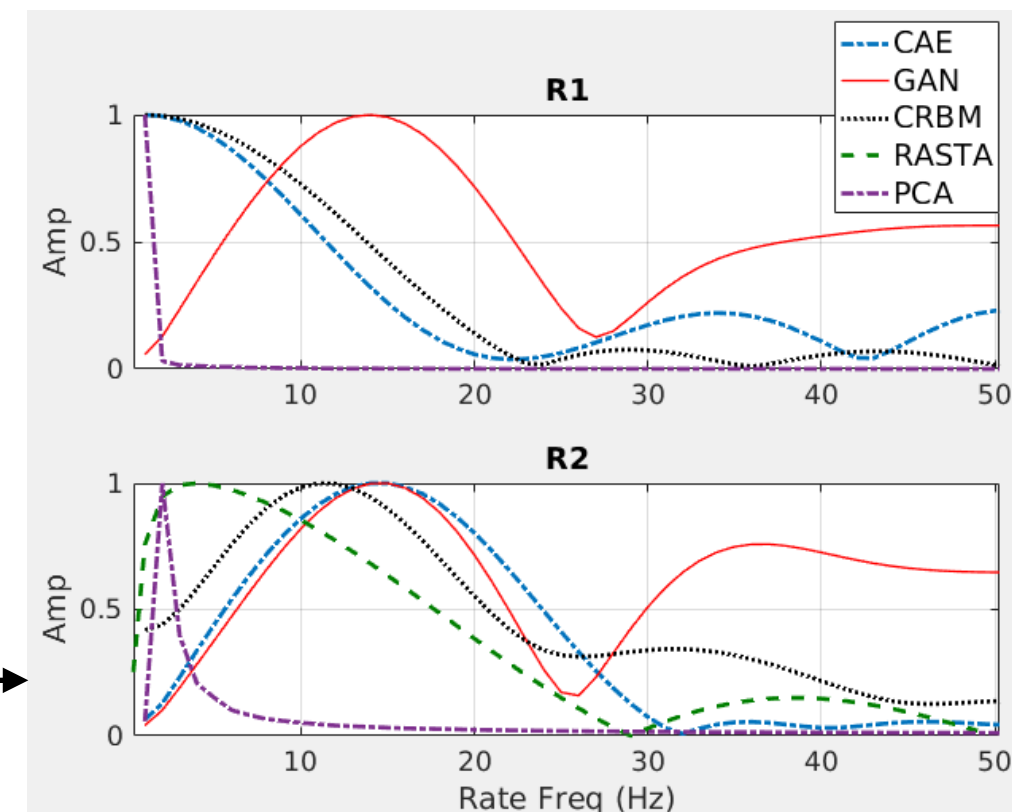
Perception based experiments on modulation filtering range : Temporal modulations within 1 – 16 Hz are crucial for speech intelligibility

Filter Selection

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- Filter selection - based on average hidden activation probability $p(\mathbf{h} = 1|\mathbf{v})$ for each of the filter obtained [1]
- With validation data spectrograms in forward pass of the trained model (CRBM, CAE, cGAN)
- R2 (bandpass rate) has higher average activations compared to other rate filters
- S1 (lowpass scale)
- Empirical validation through ASR (Word error rate)
- Features with R2-S1, R2-S2 (as parallel streams)

Selecting bandpass
rate filter



Perception based experiments: Temporal modulations within 1 – 16 Hz, spectral modulations within 0 – 4 cycles/mel, are crucial for speech intelligibility

Experiment Details

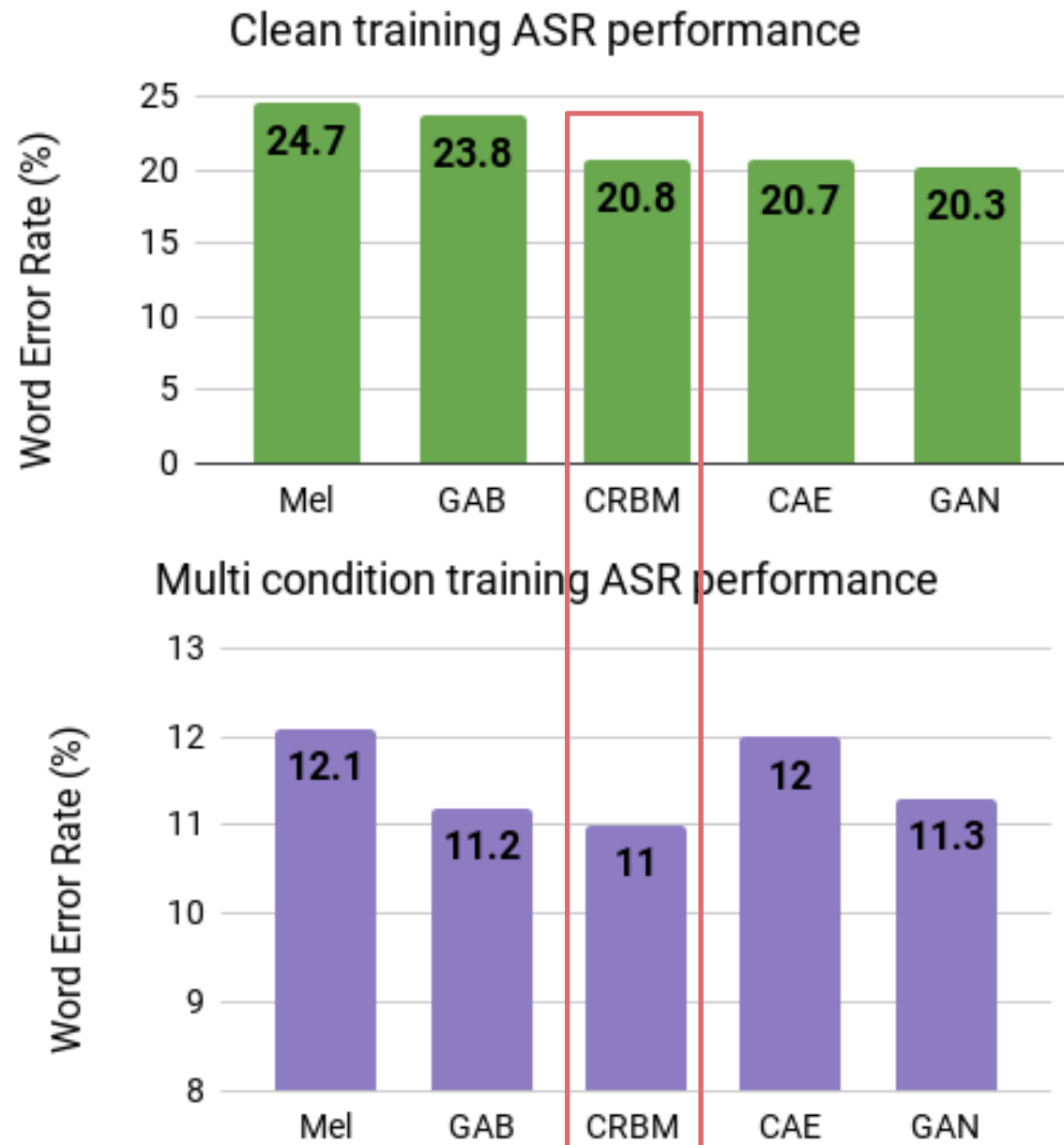
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- Feature extraction - spectrograms filtered using selected filters are fed as features to ASR
- **Automatic speech recognition (ASR)** to understand the benefit of the learnt representations
- Comparison with other features:
 - Log mel filterbank energy features (Mel)
 - Gabor features (GAB) [1]
- Wall Street Journal (WSJ) Aurora-4 speech corpus
 - Train data – 15 hours from 84 speakers:
 - clean training
 - multi-condition training (clean & noisy) - street, train, car, babble, restaurant, and airport
 - Test data – 5 hours from 8 speakers (**14** different clean and noisy conditions)
- Acoustic Model - 4 layer DNN with sigmoid non-linearity
- Our metric of ASR performance : Average of all 14 test conditions

ASR Results

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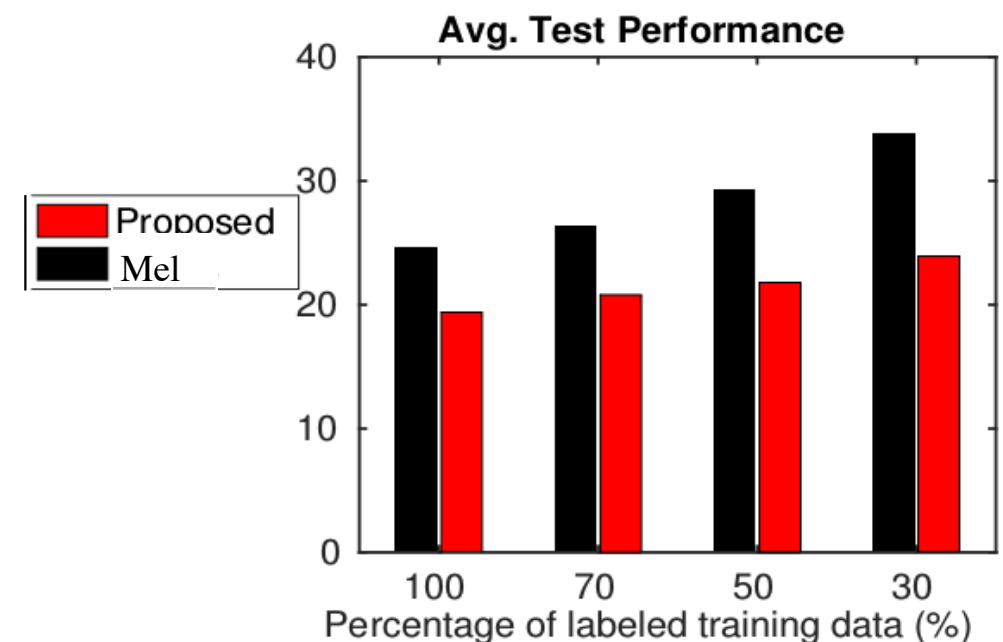
Aurora-4 - Additive noise



Average relative improvement of about 20% in clean training, 11% in multi cond. training

Semi-supervised training

- Reducing the amount of labeled training data for training DNN
- Trained DNN with 100%, 70%, 50% and 30% of the available labeled data
- Filter are learnt using 100% of unlabelled training data



With 30% of training data, average relative improvement of 26% in clean training

2-D Convolutional RBM

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- Input is 2-D spectrogram patch, filters are 2-D kernels $\longrightarrow$ To jointly learn spectro-temporal characteristics

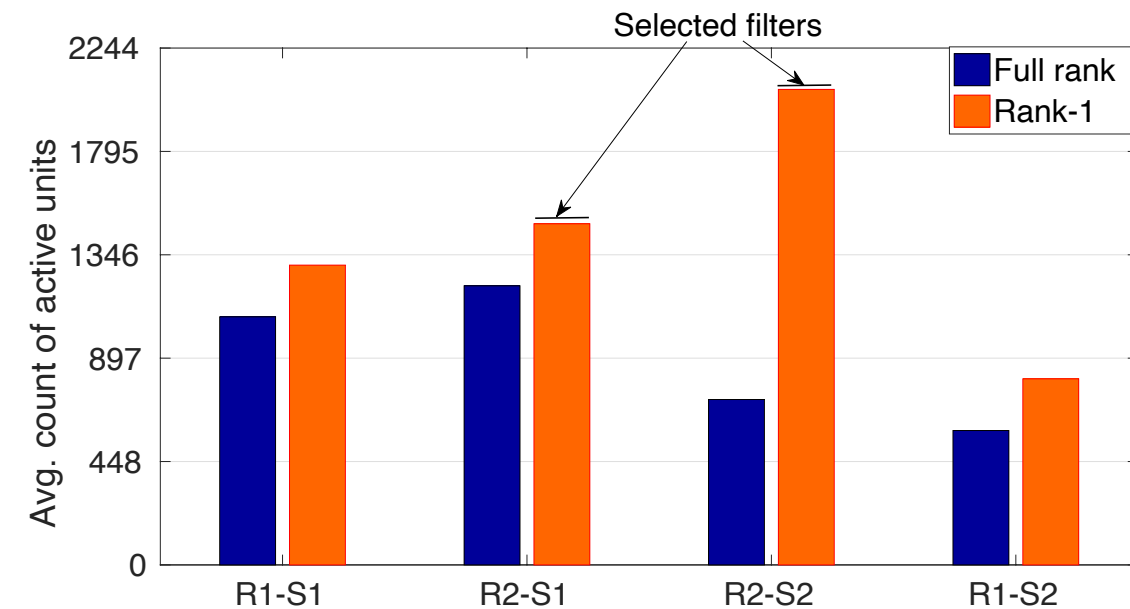
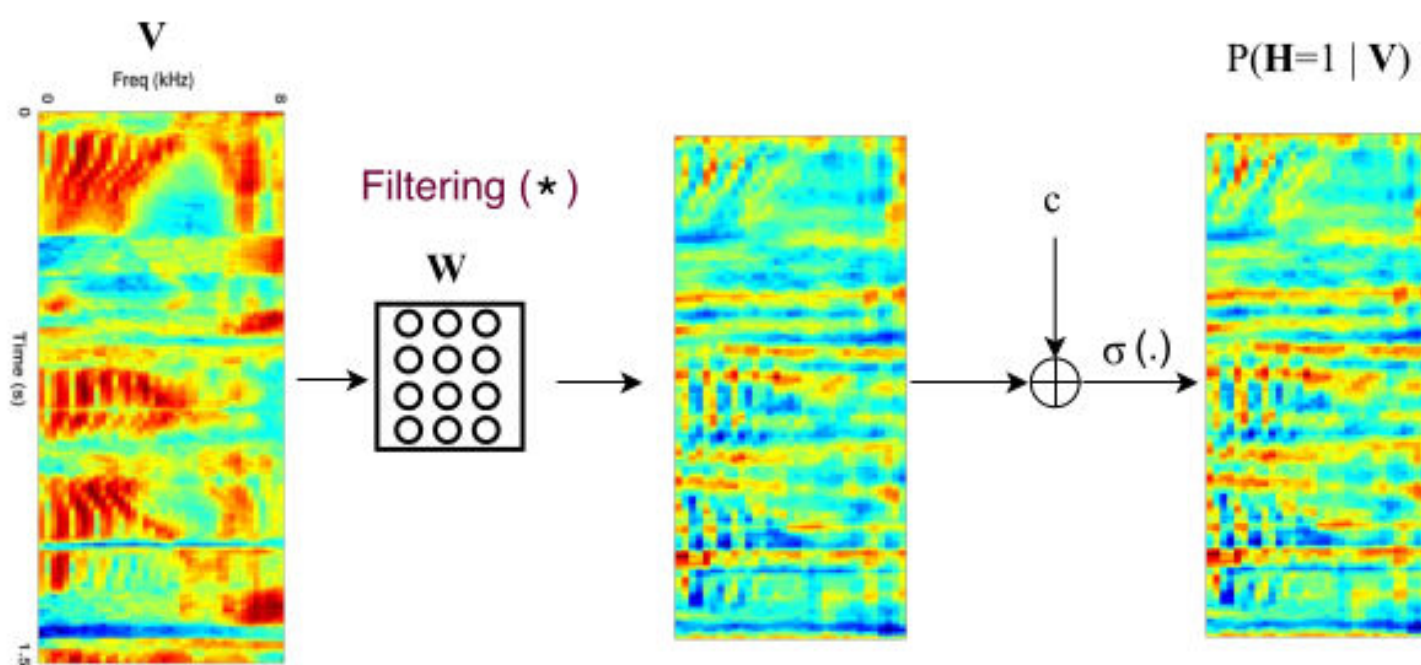
- We define $\mathbf{W}$ as the outer product of 1-D rate filter $\mathbf{r}$ and scale filter $\mathbf{s}$, i.e., $\mathbf{W} = \mathbf{r}\mathbf{s}^\top$.

- Hence the filter update equation becomes:

$$\mathbf{r}' = \mathbf{r} + \eta(\Delta_{\mathbf{r}} J); \quad \mathbf{s}' = \mathbf{s} + \eta(\Delta_{\mathbf{s}} J)$$

Rank constraint - to learn separable spectro-temporal filters, can still use residual approach

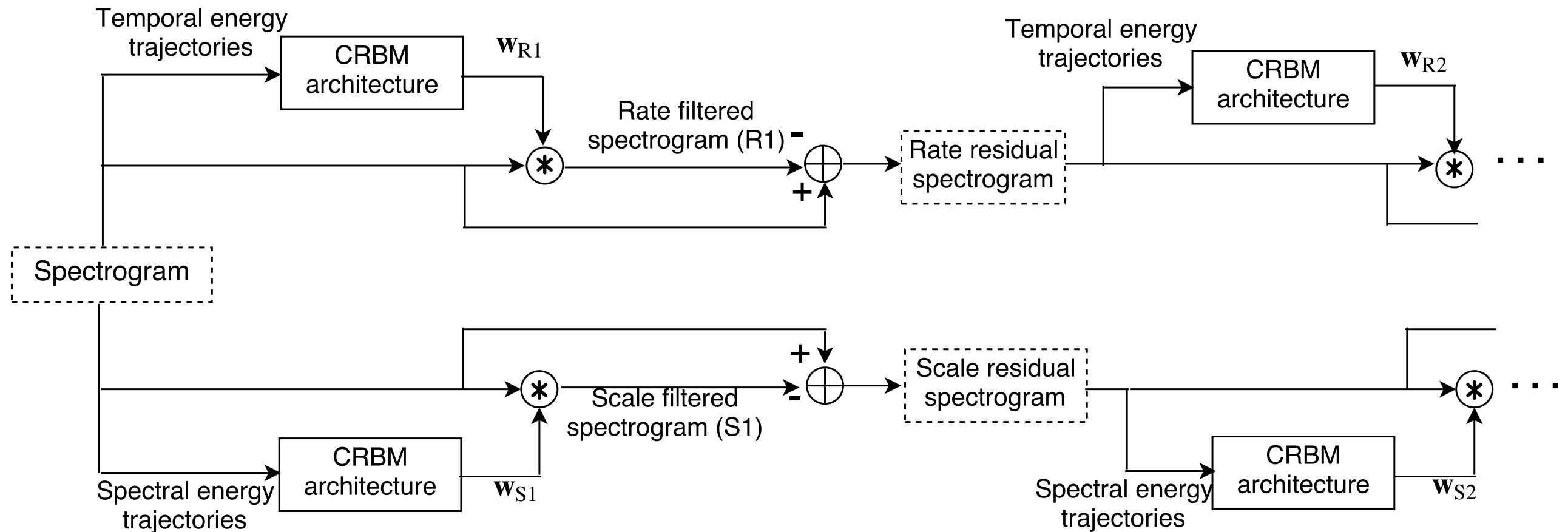
- The 2-D filter $\mathbf{W}$ is updated as $\mathbf{W}' = \mathbf{r}'\mathbf{s}'^\top$



To Learn Irredundant Filters

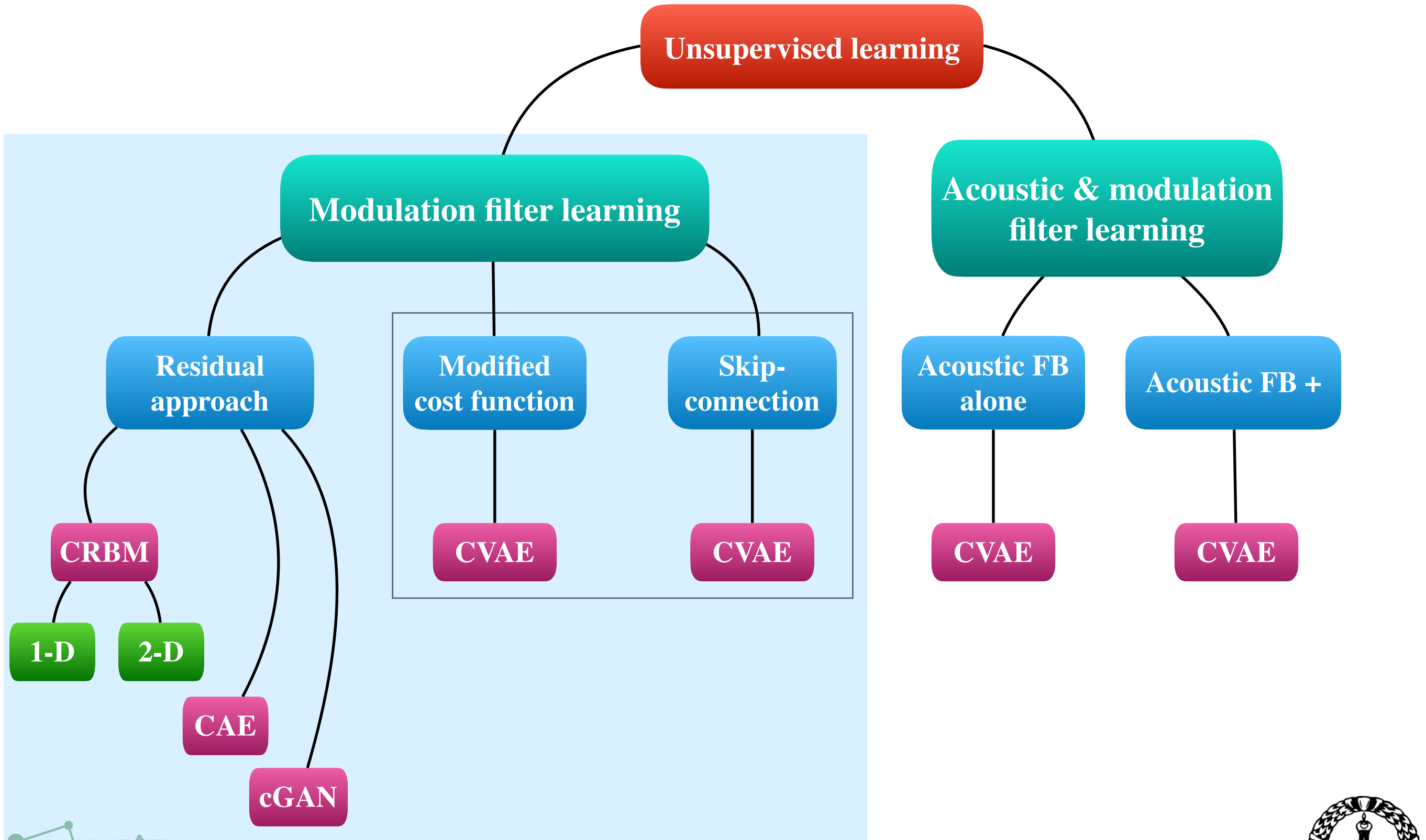
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- Residual approach



Can we avoid multiple trainings and still learn multiple irredundant filters?

Part I: Unsupervised Learning (cont...)

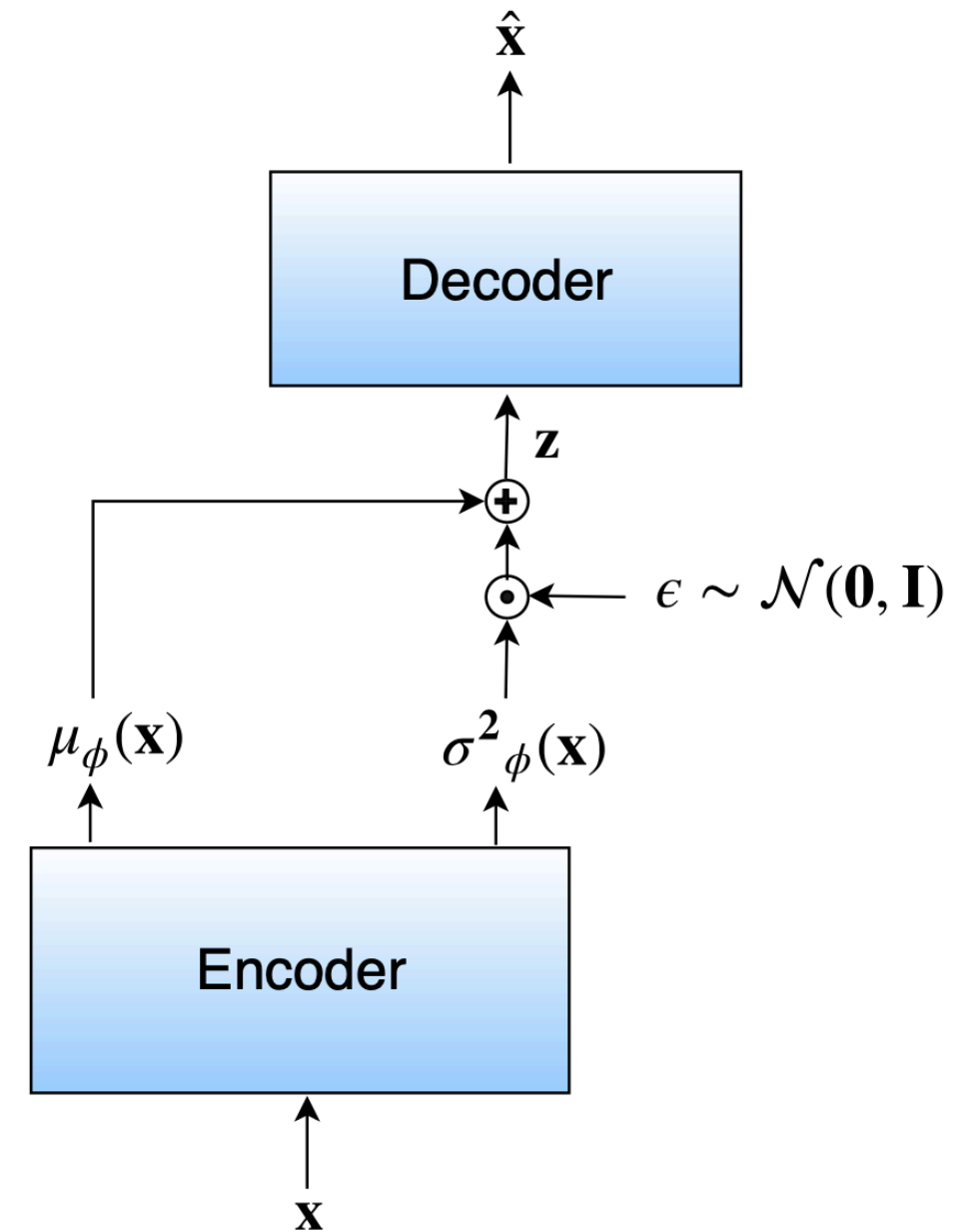


Variational Autoencoder (VAE)

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- VAE - samples of latent representation can be drawn from a standard normal distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$
- Encoder estimates the parameters of the latent data distribution as $q_\phi(\mathbf{z}|\mathbf{x})$
 - approximation of the posterior distribution $p_\theta(\mathbf{z}|\mathbf{x})$
- Trained with maximising the 'variational' lower bound:

$$\mathcal{L}(\theta, \phi; \mathbf{x}) = \underbrace{-D_{KL}[q_\phi(\mathbf{z}|\mathbf{x})||p(\mathbf{z})]}_{\text{Latent loss}} + E_{\mathbf{z}|\mathbf{x} \sim q_\phi} [\log p_\theta(\mathbf{x}|\mathbf{z})]$$



Modified Cost Function Framework

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- Convolutional VAE (CVAE)
- No. of 2-D rank-1 filters in a layer = 2
- Modified Loss function in convolutional variational autoencoder (CVAE)

$$E_{Total} = \alpha E_{MSE} + \beta E_{Latent} + \gamma(E_{fr} + E_{fs}) + \delta E_{enc}$$

$$E_{MSE}(\mathbf{x}, \hat{\mathbf{x}}) = ||\mathbf{x} - \hat{\mathbf{x}}||_2^2$$

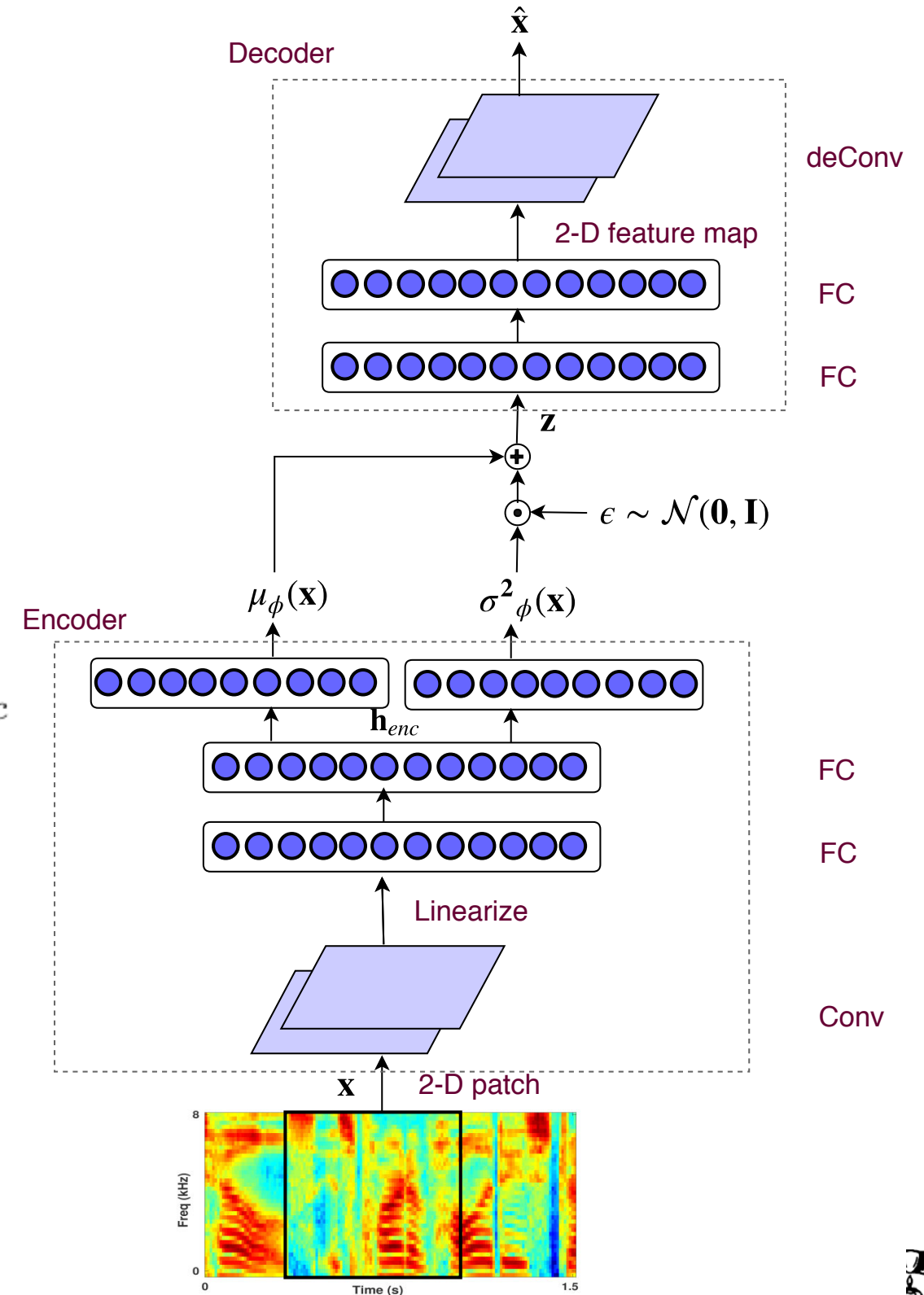
$$E_{Latent} = D_{KL}(q_{\phi}(\mathbf{z}|\mathbf{x}), p(\mathbf{z}))$$

$$E_{fr}(\mathbf{r}_1, \mathbf{r}_2) = ||\mathbf{r}_1 * \mathbf{r}_2||_2^2$$

$$E_{fs}(\mathbf{s}_1, \mathbf{s}_2) = ||\mathbf{s}_1 * \mathbf{s}_2||_2^2$$

$$E_{enc}(\mathbf{h}_{enc}) = ||\mathbf{h}_{enc}||_1$$

To obtain non-overlapping filters

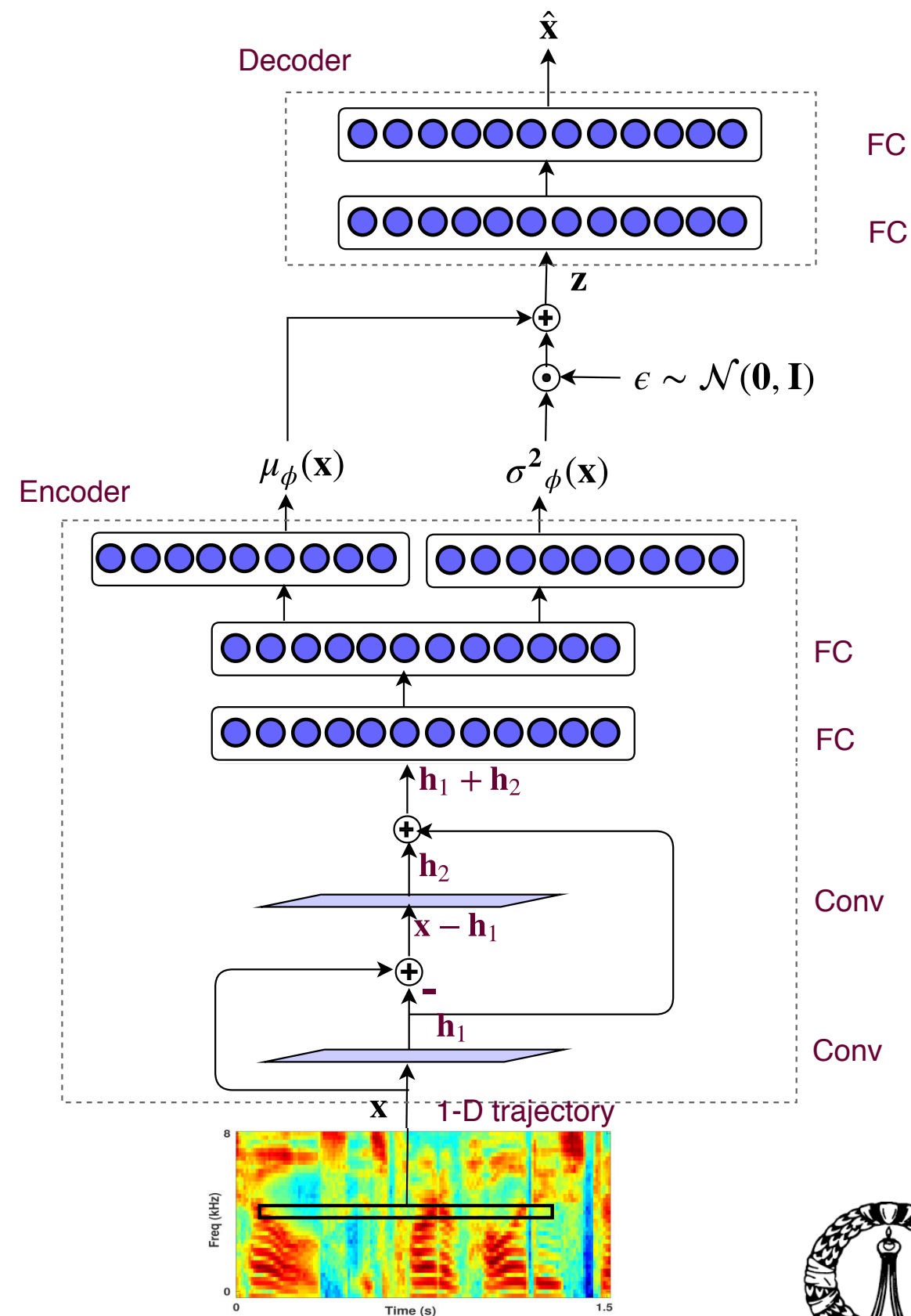


Skip-Connection Framework

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- The 'skip' architecture serves two purposes:
 - to find the residual of the input with some filtered representation
 - to combine the two different filtered representations
- Some way - imitates previous residual approach
- With x as the input temporal trajectory, $h_1 = x * r_1$
- The residual, i.e. $(x - h_1)$ is fed to next convolution layer

$$h_2 = (x - h_1) * r_2$$
- The output of next layer is $\tanh(h_1 + h_2)$
- The two representations are then added and fed to the next layer of CVAE



ASR Performance

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Summary and comparison of all unsupervised approaches
for the ASR task on Aurora-4 dataset

Model	Cost-function	Method of learning	Type of filtering	ASR (WER %)	
				Clean	Multi
Baseline	-	-	-	24.7	12.1
CRBM 1-D	Boltzmann likelihood	Residual external	Rate	20.8	11.0
CAE 1-D	Mean square error			20.7	12.0
cGAN 1-D	Adversarial loss			20.3	11.3
CRBM 1-D	Boltzmann likelihood	Residual external	Rate-scale	19.4	10.8
CVAE-skip 1-D	Variational loss	Skip-connection		20.5	11.0
CRBM 2-D	Boltzmann likelihood	Residual external	Rate-scale	19.9	11.1
CVAE ModC 2-D	Variational (All 4 terms)	Modified cost function		22.1	11.2

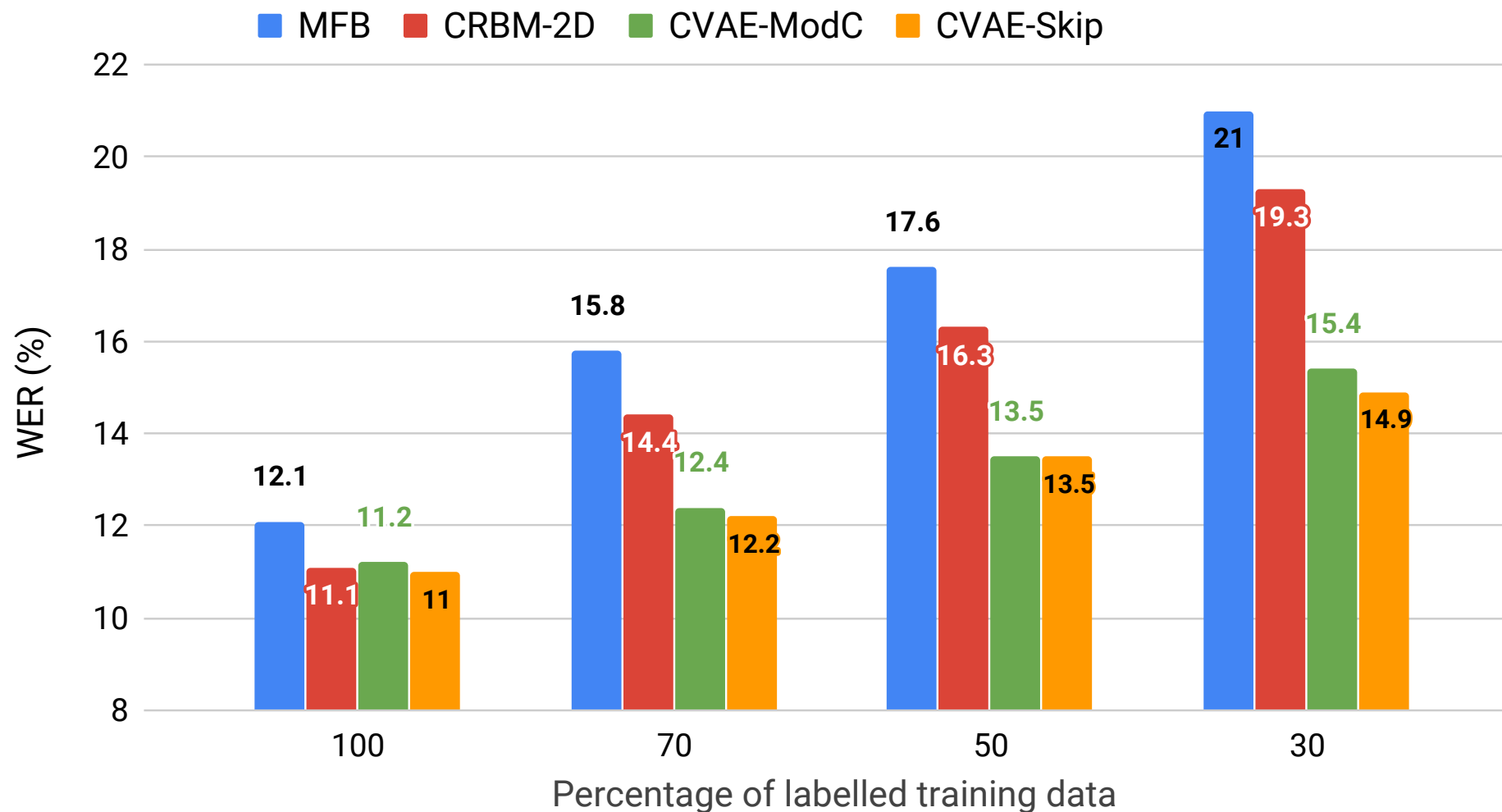
CNN-DNN for the ASR task on Aurora-4 dataset - supervised learning (64 kernels)

CNN 1-D	-	-	Rate	-	11.3
CNN 2-D	-	-	Rate-scale	-	11.0

ASR Performance

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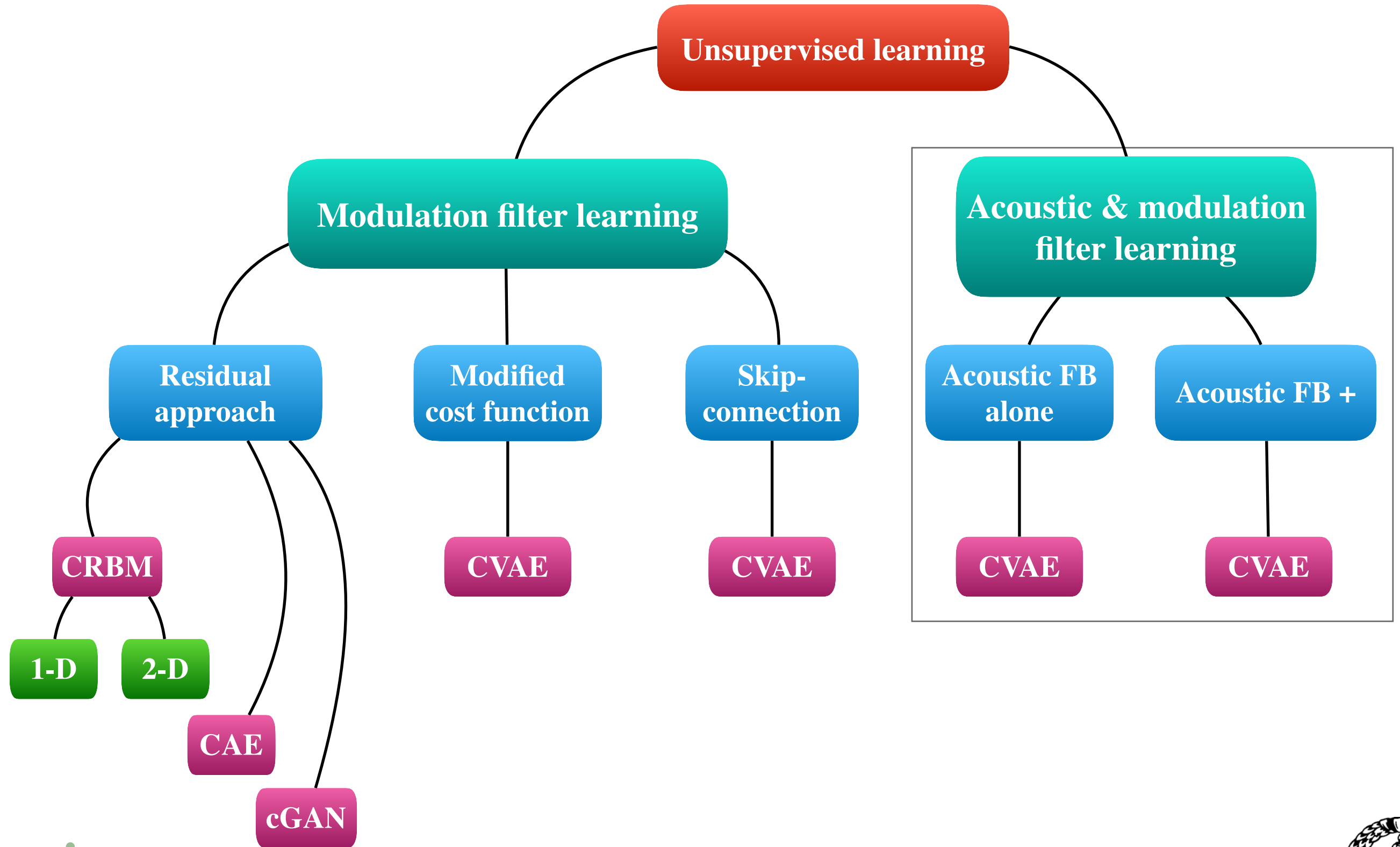
- Semi-supervised training setup (multi-condition training)



- Cross-domain ASR experiments
 - Aurora-4 - noisy data
 - REVERB - reverberant data
 - CHiME-3 - noisy + reverberant data

CVAE-ModC Filters Learned on	ASR trained and tested on		
	Aurora-4	REVERB	CHiME-3
Aurora-4	11.2	16.2	15.0
REVERB	11.0	16.1	15.2
CHiME-3	11.2	16.0	15.3

Part I: Unsupervised Learning (cont...)



Representation Learning From Raw Waveform

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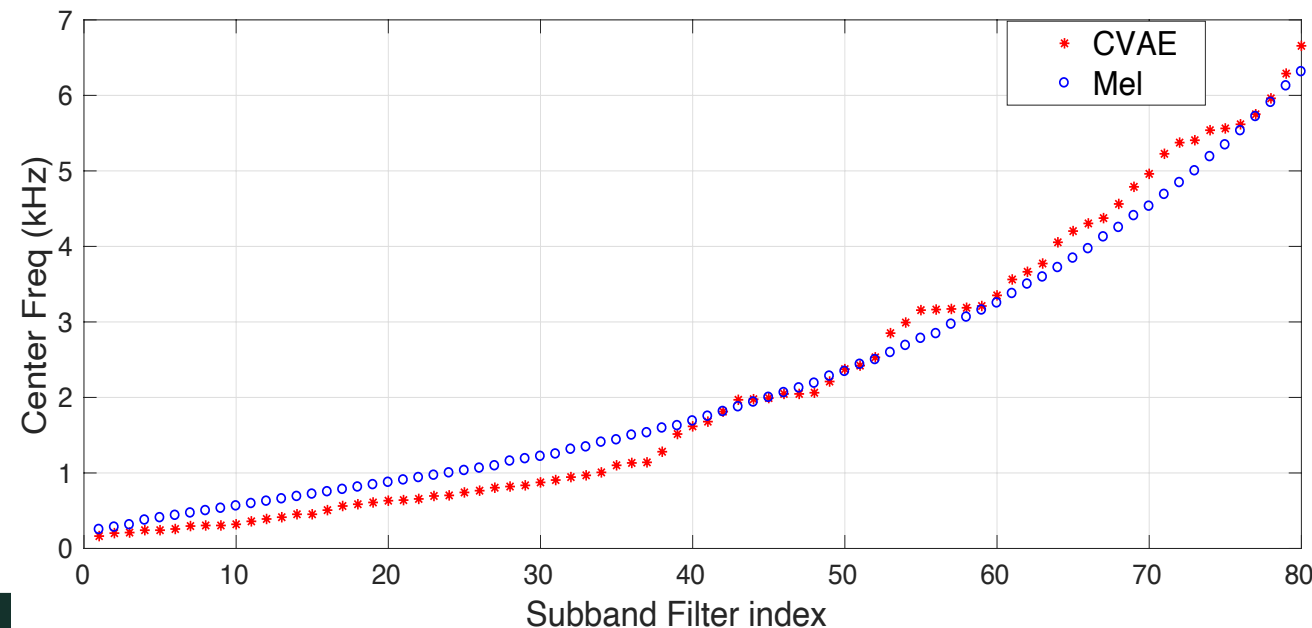
- Step-1: Representation learning from raw waveform
- Step-2: Modulation filtering on obtained representation
- The kernels (filters) of the convolution layer are modelled using cosine-modulated Gaussian function as:

$$w_n(t) = \cos 2\pi\mu_n t \times \exp(-t^2\mu_n^2/2)$$

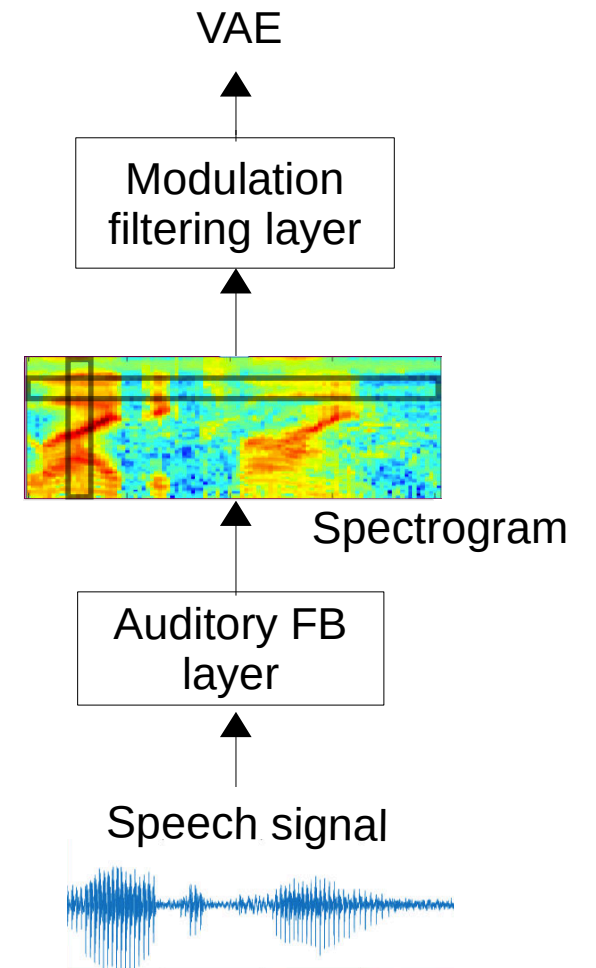
$w_n(t)$ n-th filter impulse response at time t

μ_n Freq of the n-th filter's cosine function - learnable

- Cost function as: $E_{Total} = \alpha E_{MSE} + \beta E_{MSE-acoustic} + \gamma E_{Latent}$

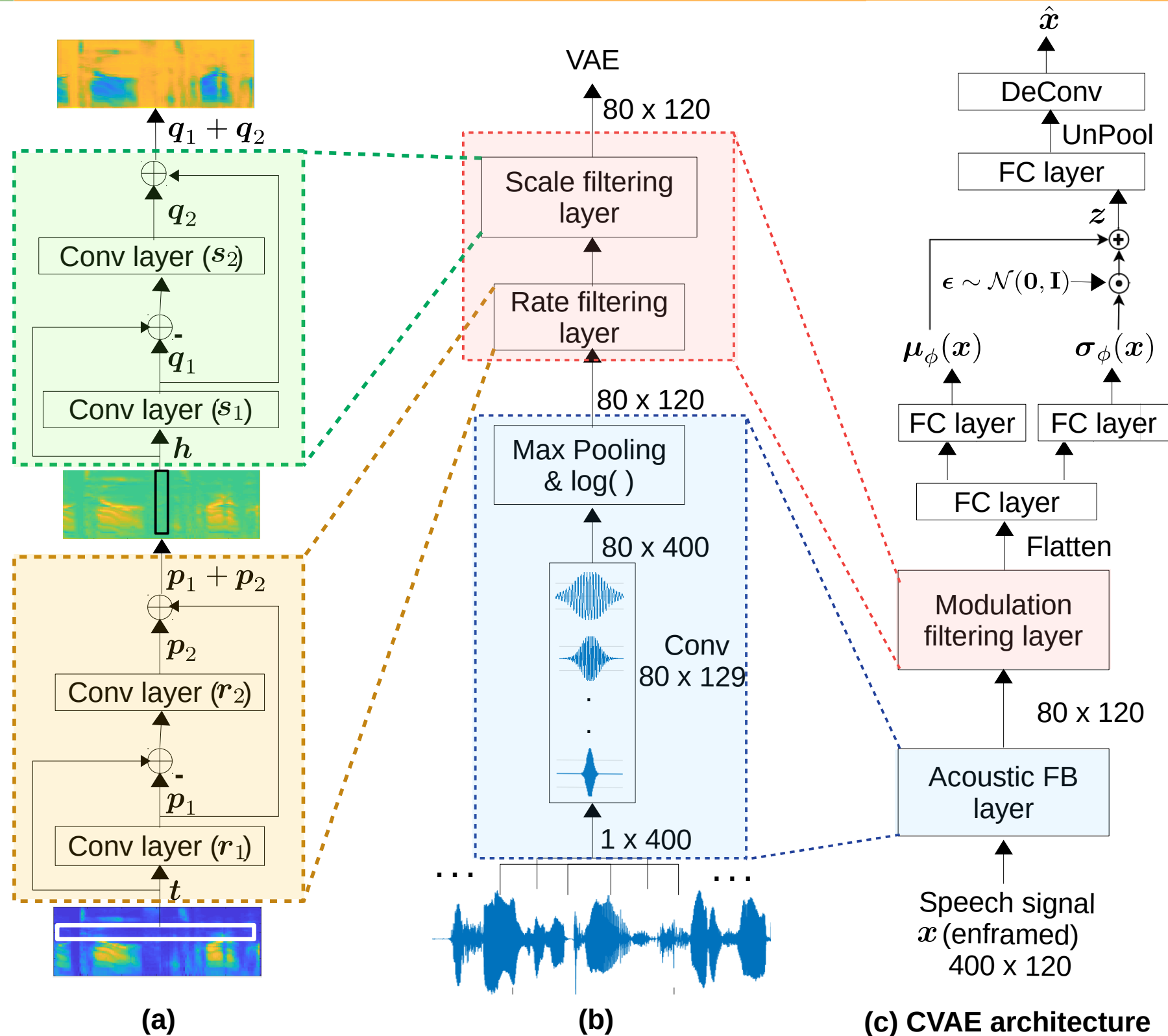


MSE between spectrogram representations in Encoder and Decoder



Acoustic FB Learning+ Modulation Filtering

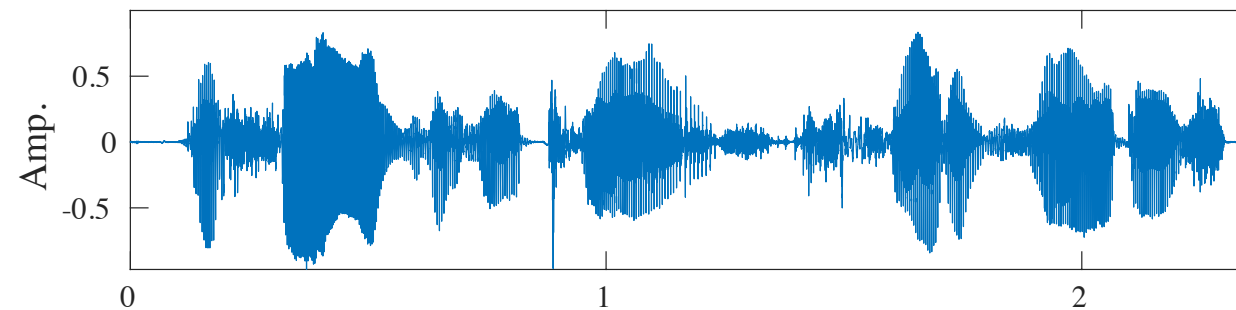
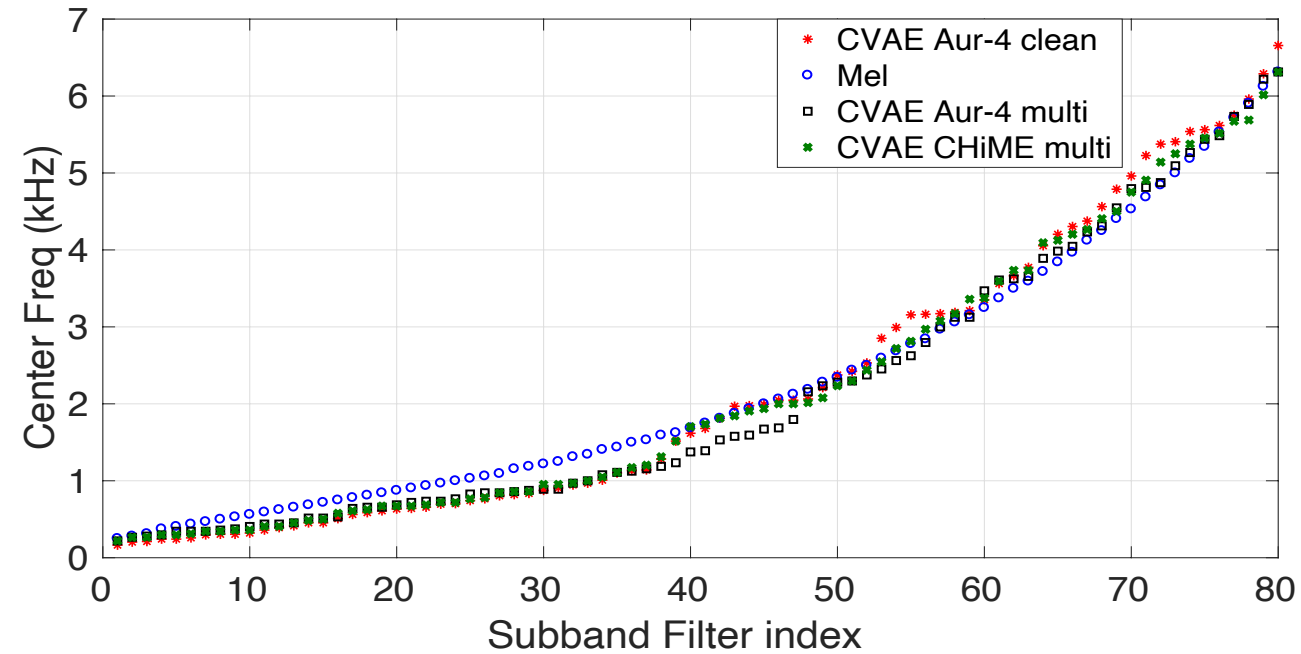
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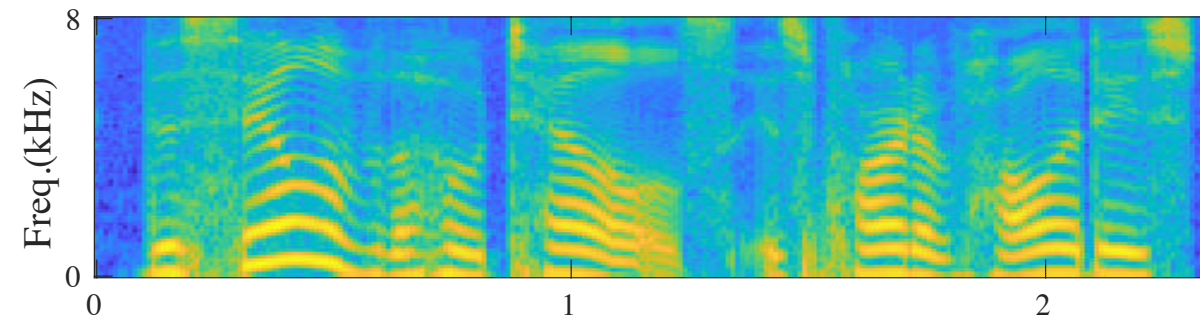
$$E_{Total} = \alpha E_{MSE} + \beta E_{MSE-acoustic} + \gamma E_{Latent}$$

Learnt Representation

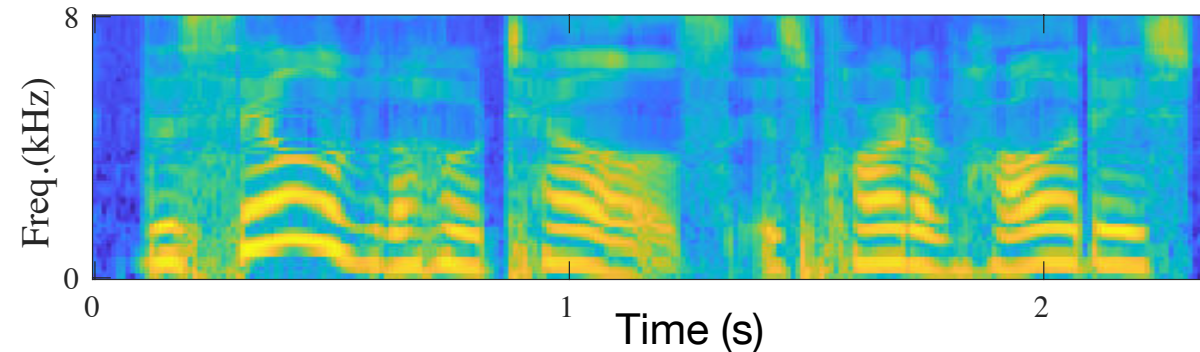
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(a) Speech waveform



(b) Mel spectrogram



(c) Learnt spectrogram

ASR Performance

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Summary for learnt acoustic filterbank learning for the ASR task on Aurora-4 dataset

Model	Cost-function	Method of learning Acoustic FB	Method of learning modulation filters	Type of filtering	ASR (WER %)	
					Clean	Multi
Baseline (Mel filterbank)	-	-	-	-	24.7	12.1
CVAE-A	Variational loss (with extra MSE	CVAE on raw	-	-	24.6	12.2
CVAE-A,M			Skip-connection	Rate-scale	20.6	11.4

Summary for discussed approaches for the ASR task on CHiME-3 dataset

Model	Cost-function	Method of learning Acoustic FB	Method of learning modulation filters	Type of filtering	ASR (WER %)	
					Multi	
Baseline	-	-	-	-	13.0	
CVAE-A	Variational loss (with extra MSE term)	CVAE on raw	-	-	12.8	
CVAE-A,M			Skip-connection	Rate-scale	11.1	

Part I: Summary

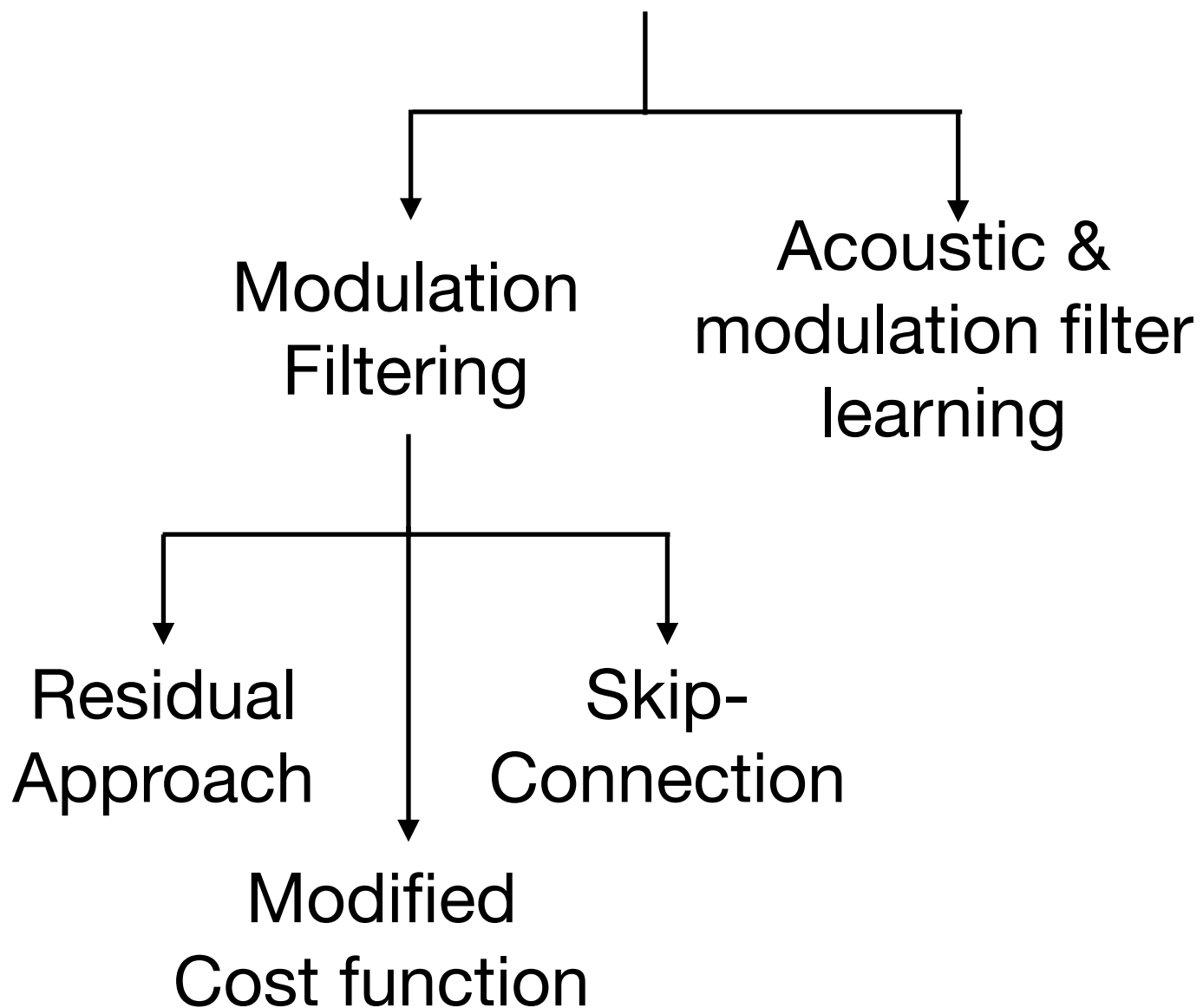
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- Part I : Unsupervised representation learning
 - Our approach - robust representations by modulation filtering method
 - Filter the spectrograms to remove irrelevant information
 - Proposed 3 different methods for learning irredundant filters - residual approach, modified cost function, and skip-connection based approach
 - Proposed and compared models for modulation filter learning - CRBM, CAE, GAN, CVAE
 - Observed improvements in ASR performance, semi-supervised training, cross-domain analysis
 - Acoustic filterbank learning from raw waveform

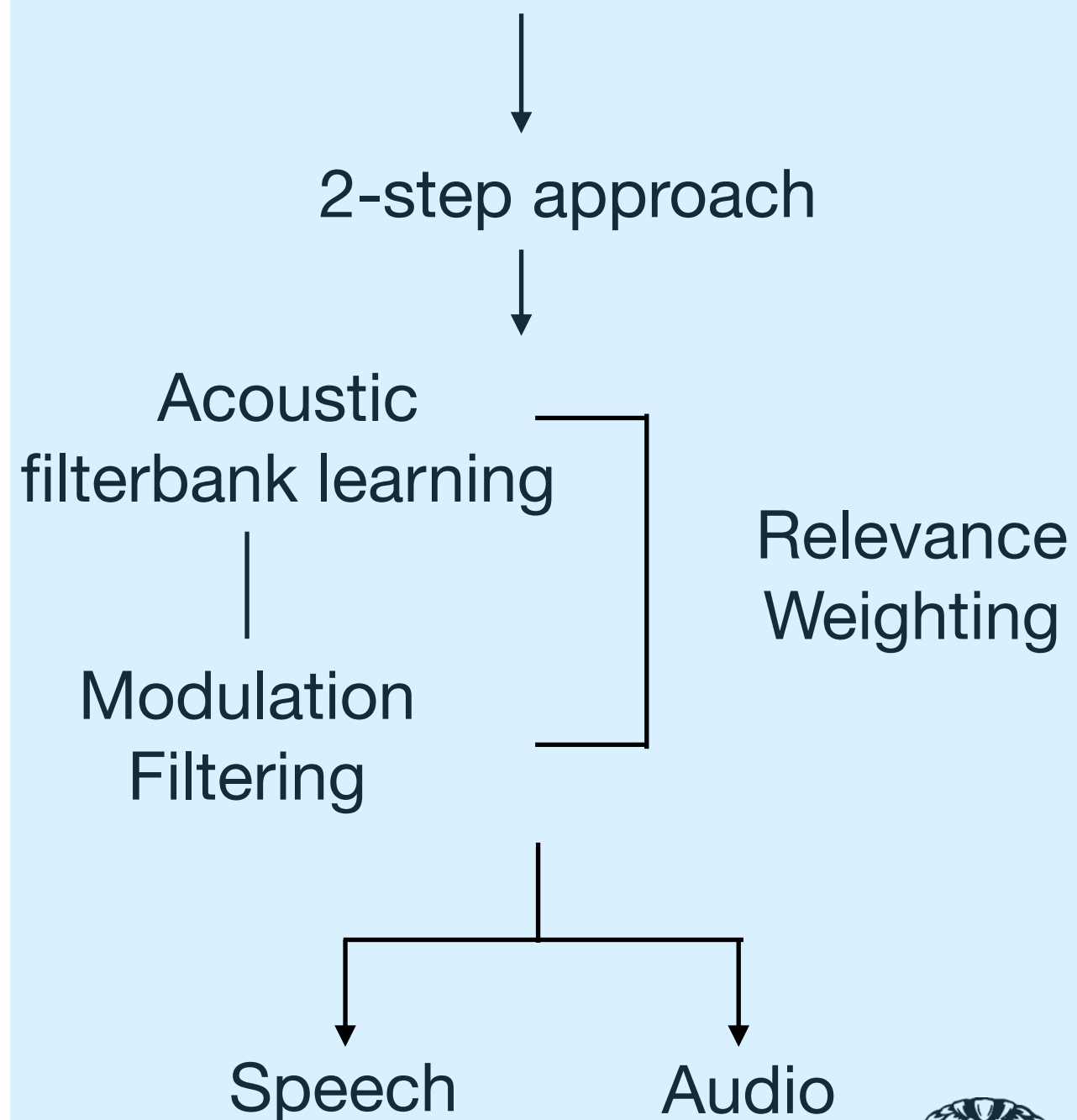
Thesis Map

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Part-I Unsupervised Learning (Speech)

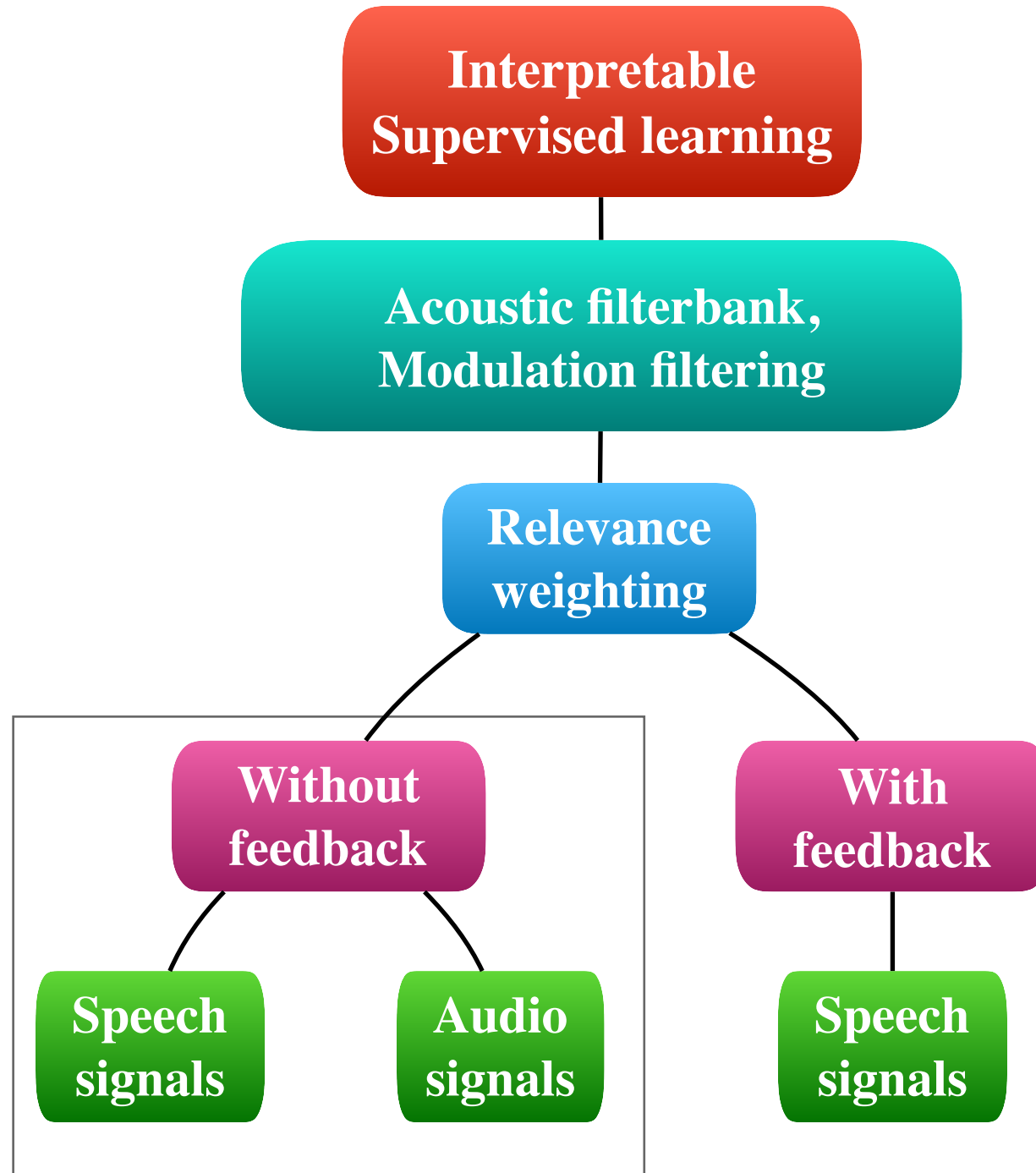


Part-II Supervised Learning



Part II: Supervised Learning

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Interpretable Representations

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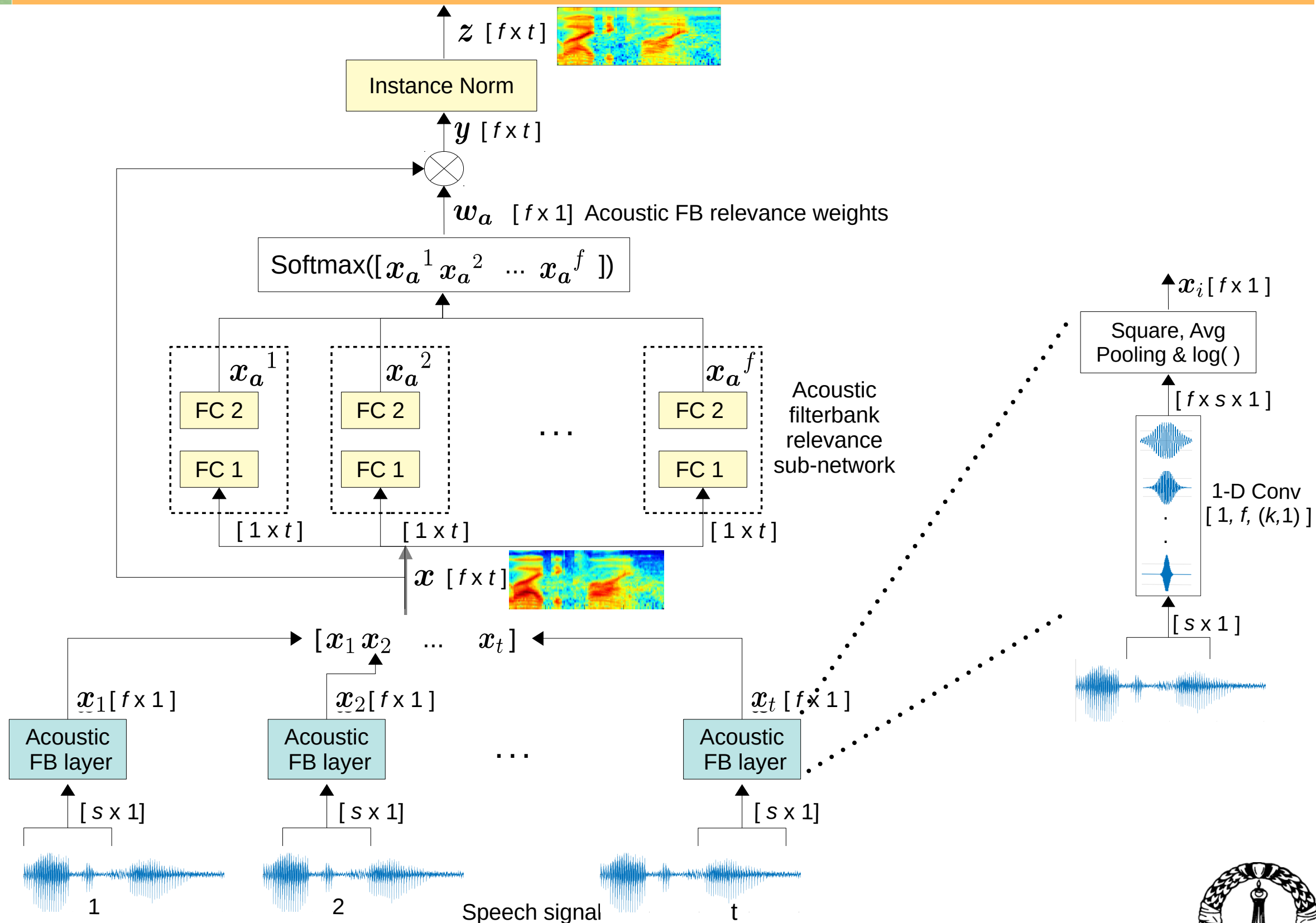
- We learn network parameters (including acoustic filterbank) in supervised fashion
- Both acoustic filterbank and modulation filterbank - parametric filter banks
- Modulation filterbank - 2D cosine-modulated Gaussian filters

Acoustic FB: $w_n(t) = \cos 2\pi\mu_n t \times \exp(-t^2\mu_n^2/2)$

Modulation FB: $w_n(t, f) = \cos 2\pi(\mu_{t_n} t \pm \mu_{f_n} f) \times \exp [(-t^2\mu_{t_n}^2) + (-f^2\mu_{f_n}^2)]$

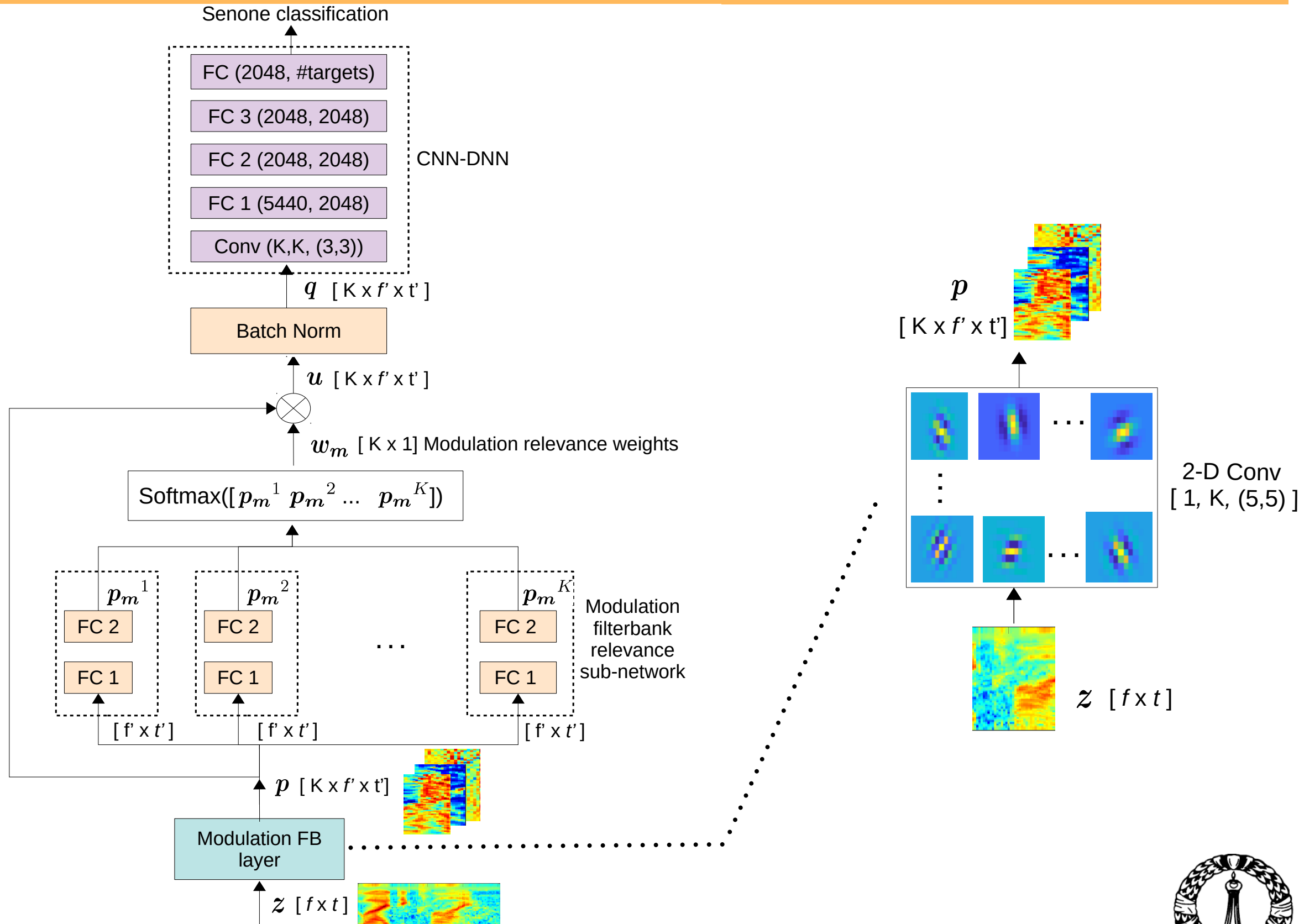
- Motivations: Interpretability of representations, explainability of network parameters
- We try to address questions using **relevance weighting** mechanism
 - Are all acoustic and modulation filters equally important for a given target?
 - Can the network learn which acoustic sub-bands/modulation characteristics are more important for the test input?

Interpretable Representation Learning



Interpretable Representation Learning

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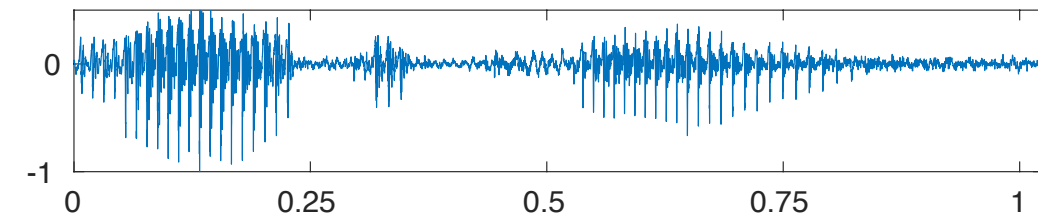
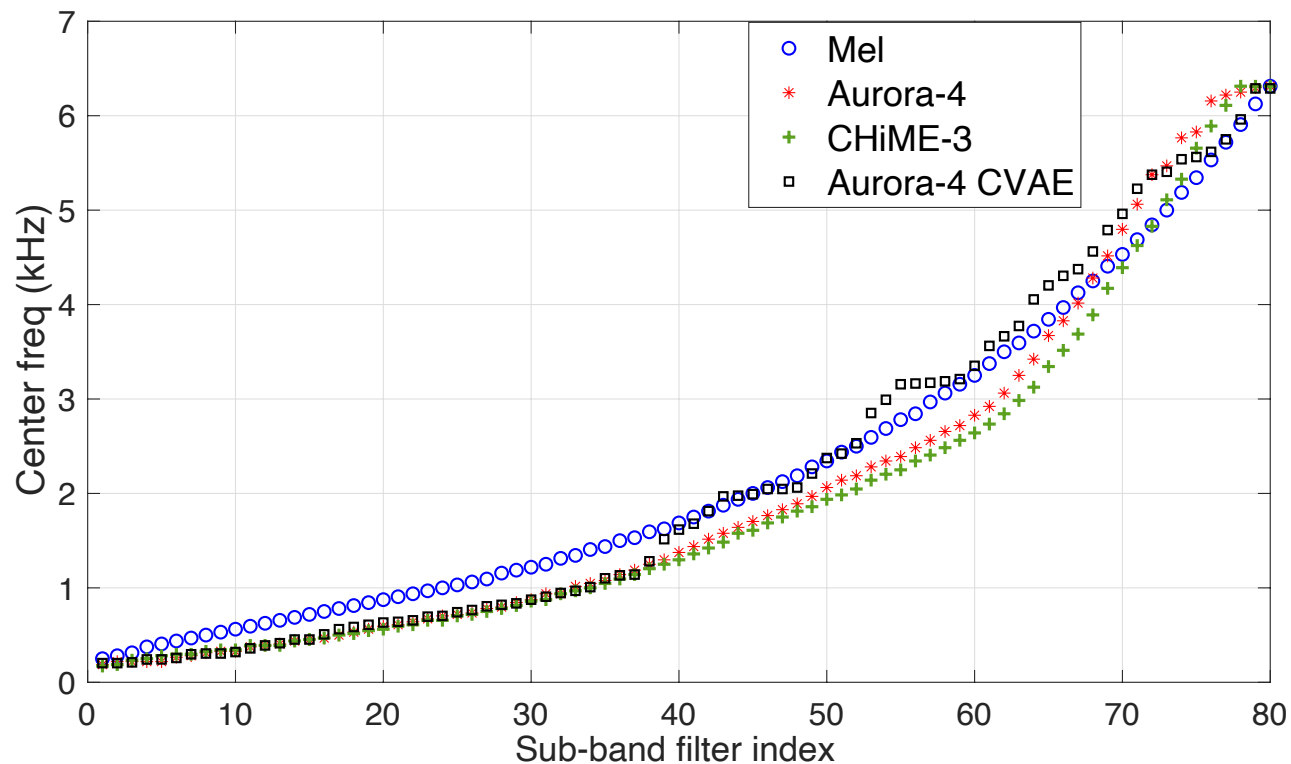


Learnt Representation

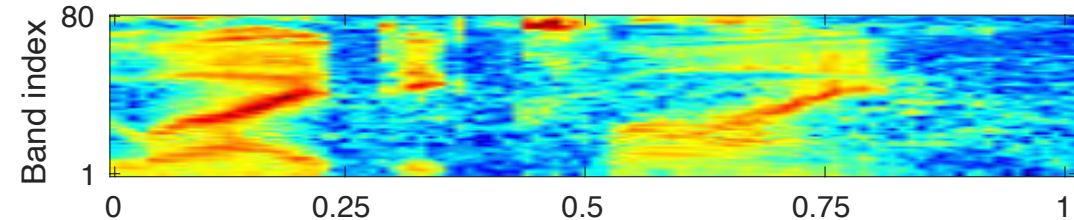
44

We begin with means learnt using CVAE in unsupervised manner

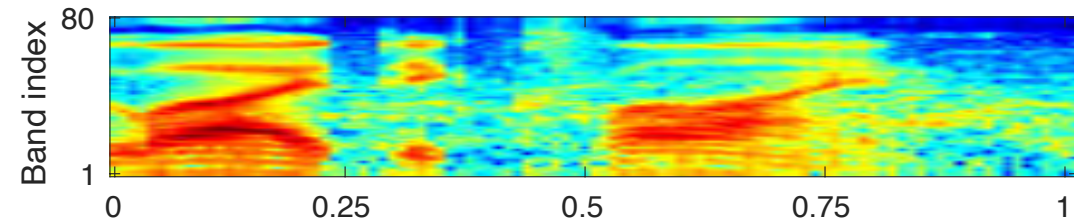
Centre frequency (μ_n) of acoustic filterbank learned using the discussed approach for Aurora-4 dataset



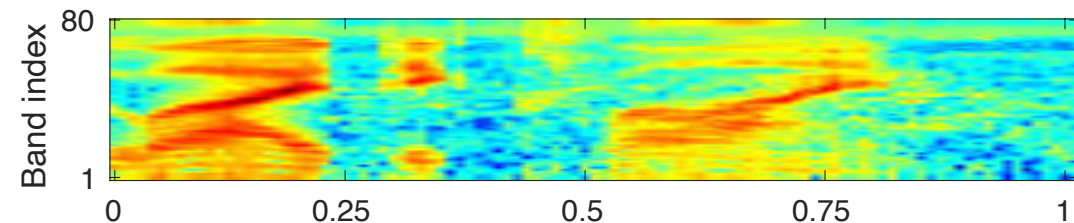
Speech waveform



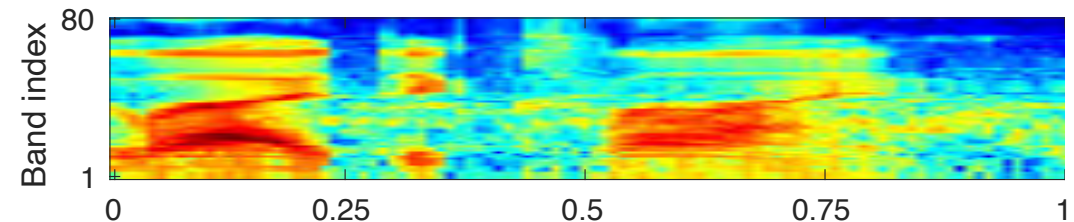
Mel Spectrogram



Learnt Spectrogram



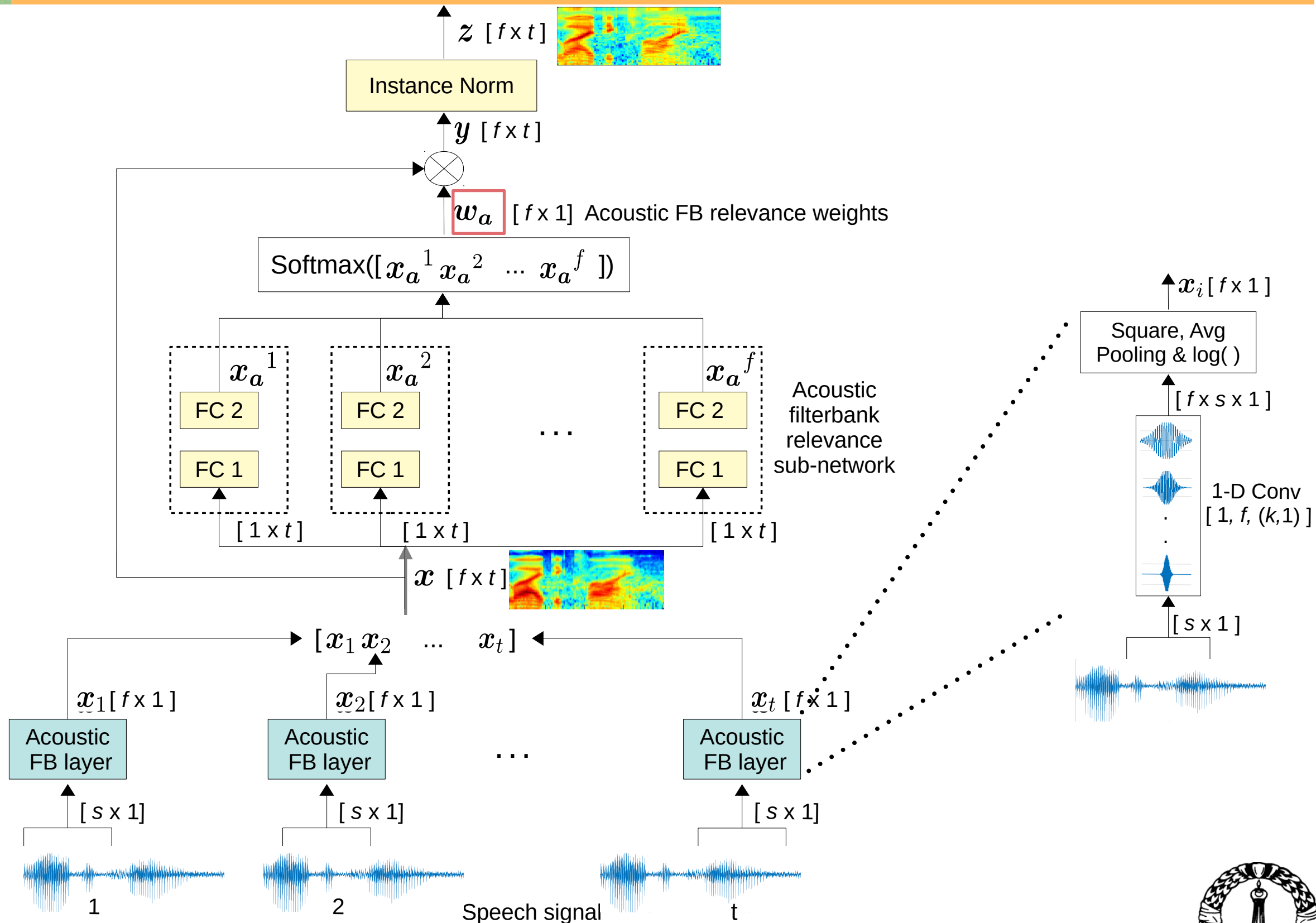
Learnt Spectrogram + weighting



Learnt Spectrogram using CVAE

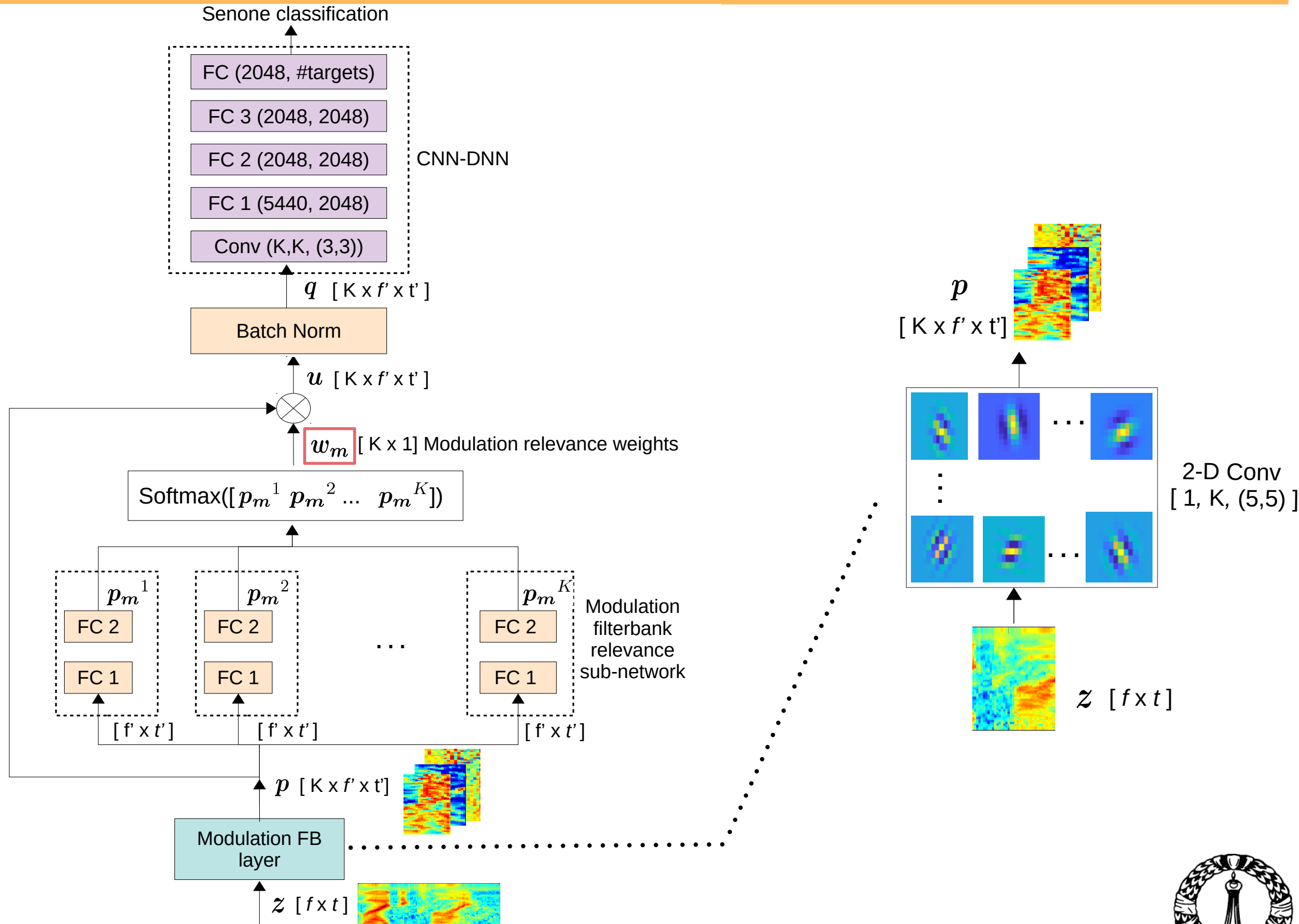
Time (s)

Interpretable Representation Learning



Interpretable Representation Learning

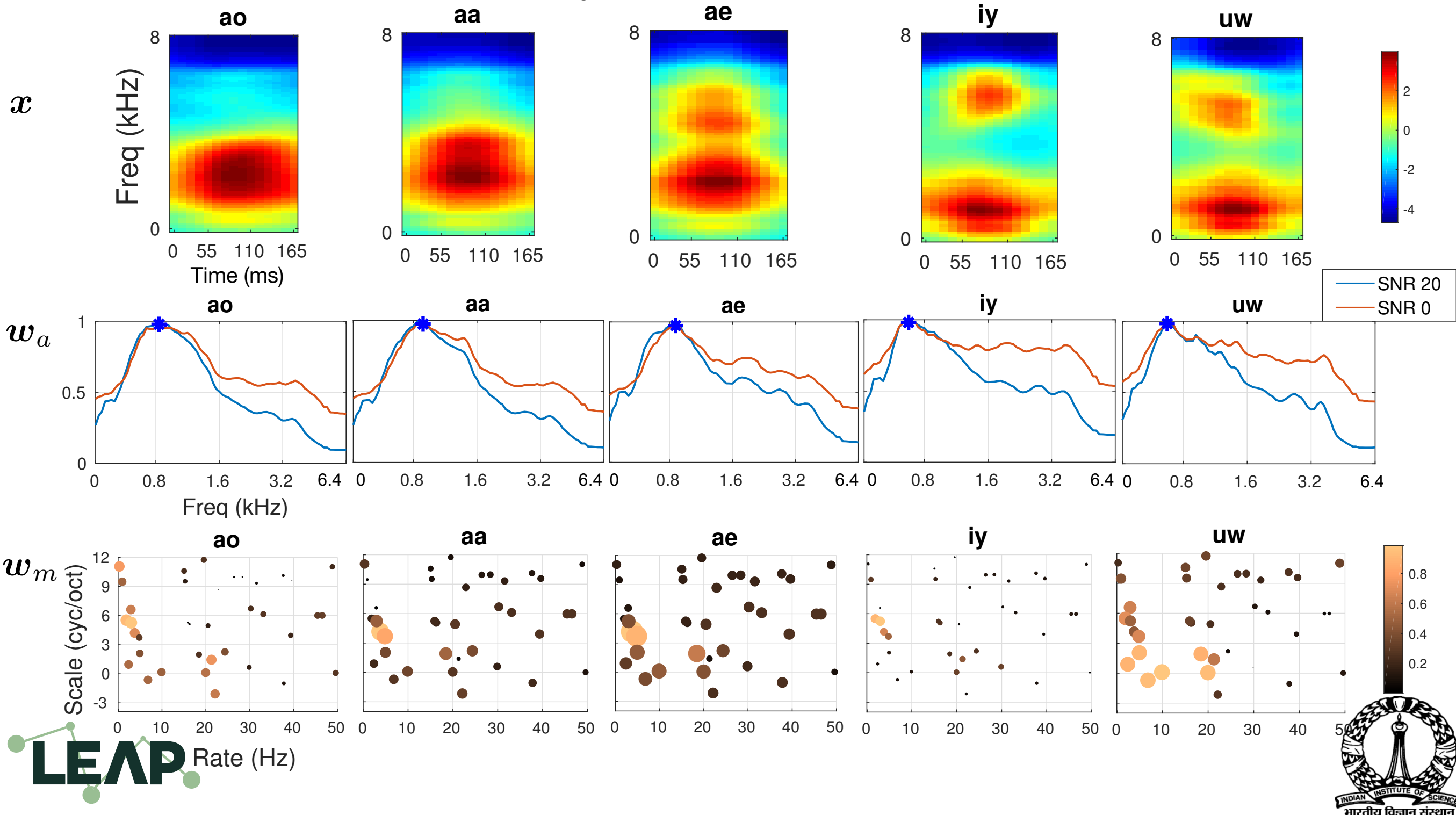
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Analysis of Representations, Weights

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- Noisy TIMIT - vowels {/ao/, /aa/, /ae/, /iy/, /uw/} - arranged according to place of articulation- back to mid to front
- We observe $\mathbf{x}$, $\mathbf{w}_a$, and $\mathbf{w}_m$ (no retraining, the Aurora-4 model is used for forward pass)



Impact on ASR

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Effect of different components of the discussed model for the ASR task on Aurora-4 dataset

Features	ASR (WER in %)
MFB (Baseline with mel)	10.7
A (Learnt acoustic FB)	10.7
MFB-R (Acoustic relevance on MFB)	10.8
A-R (Acoustic relevance on A)	10.0
A-R,M-R (Acoustic & Mod. Relevance on A)	9.6
S - Sinc filterbank [Ravenelli et. al., 2018]	10.8
S-R,M-R (Acoustic & Mod. Relevance on S)	10.0

Unsupervised vs. Supervised learning of acoustic filter bank with [A-R,M-R] configuration

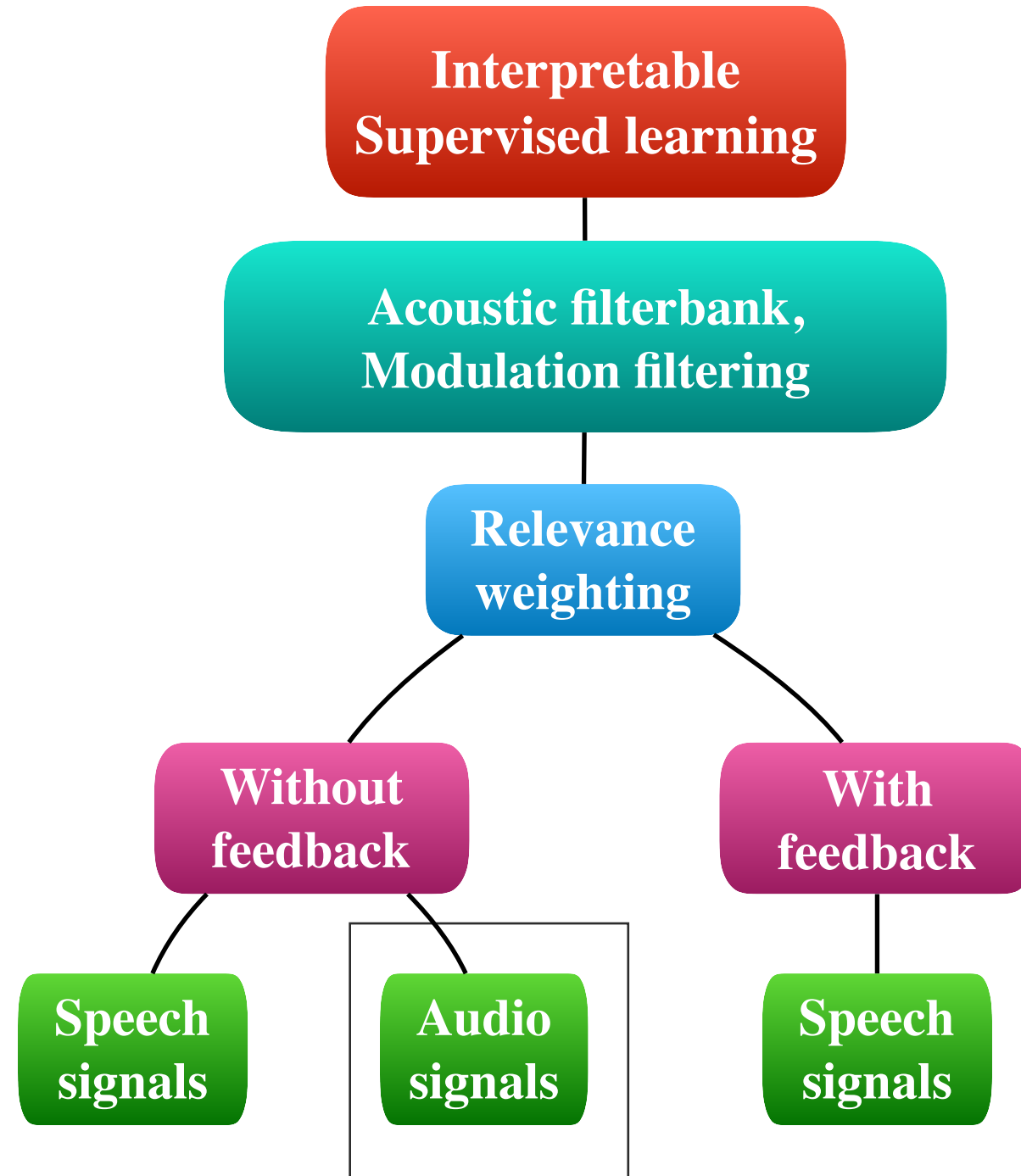
Features	ASR
Unsupervised Initialization (& no fine tuning)	10.9
Unsupervised Initialization- CVAE & supervised fine tuning	9.6
Random Initialization & supervised fine tuning	9.9
Random Initialization & No fine tuning	11.2

Cross-domain ASR experiments

[A-R,M-R] Learned on	ASR trained and tested on		
	Aurora-4	CHiME-3	VOiCES
Aurora-4	9.6	14.3	57.6
CHiME-3	9.7	14.2	59.0
VOiCES	9.9	14.4	57.2

Part II: Supervised Learning (cont)

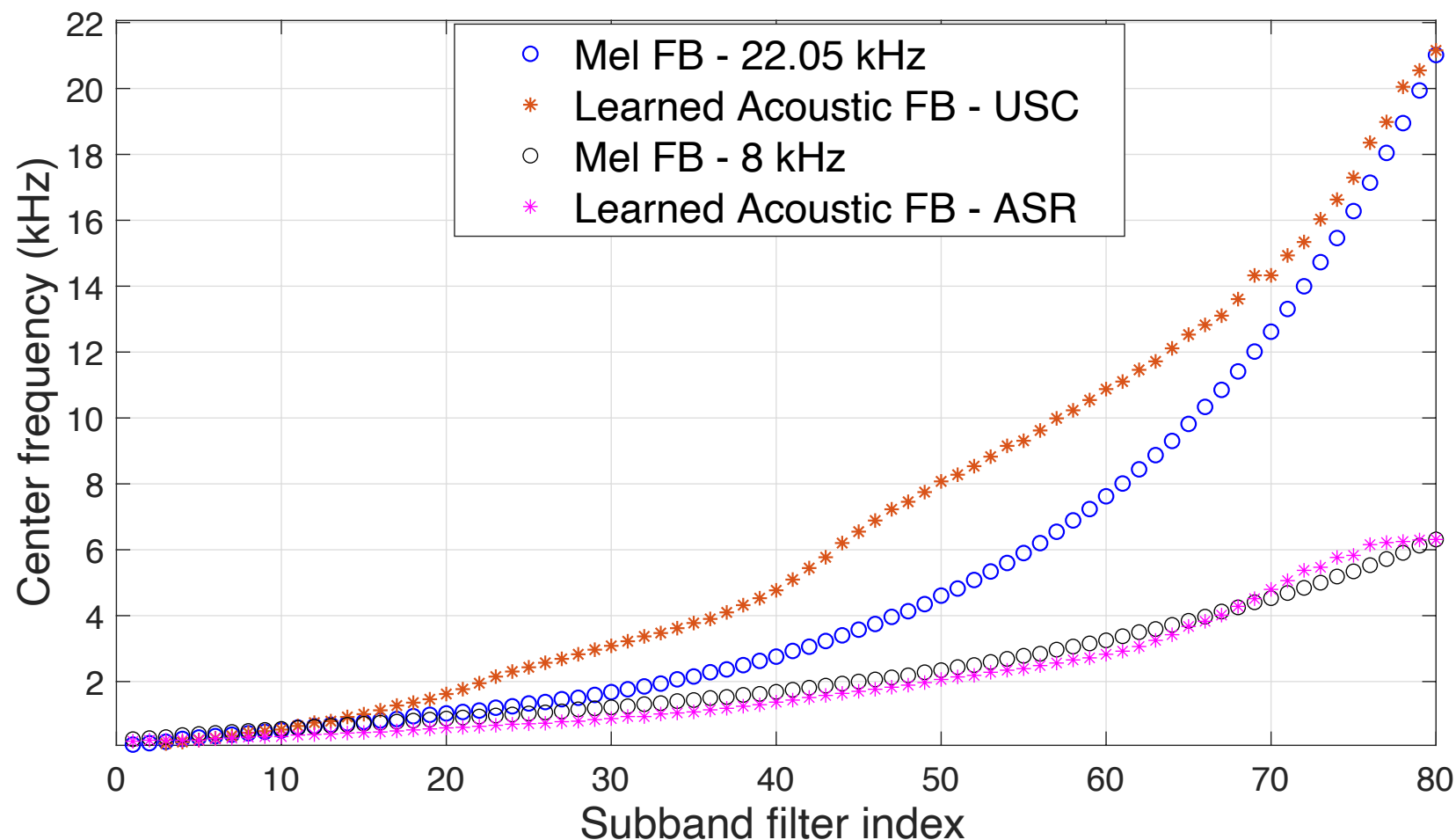
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Extending to Audio Sounds

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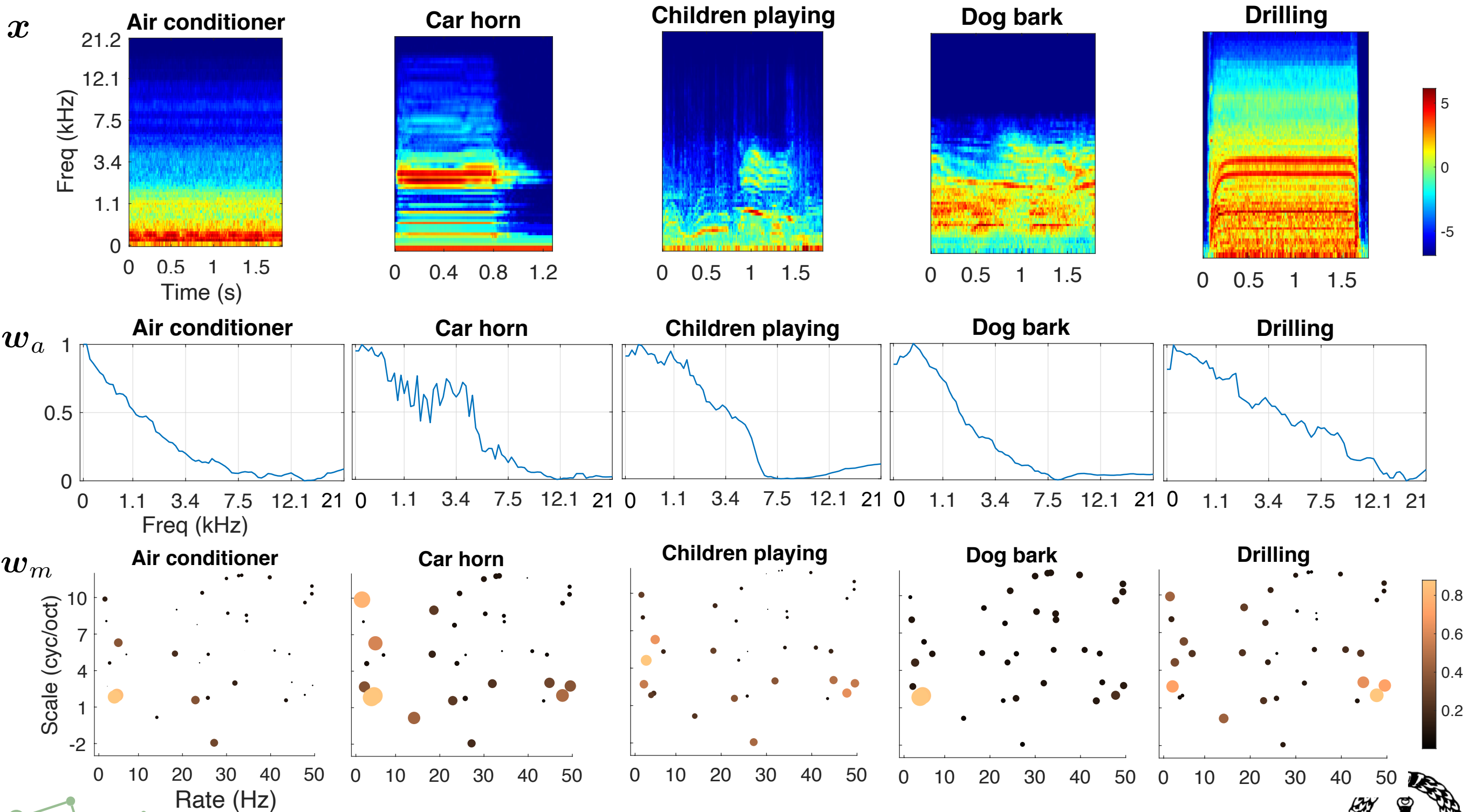
- Urban sound classification (USC) - same 2-stage approach with relevance weighting
- UrbanSound8K dataset - real-time sound excerpts, sampled at 44.1kHz
- 10 types of urban sounds - air conditioner (AI), car horn (CA), children playing (CH), dog bark (DO), drilling (DR), engine idling (EN), gun shot (GU), jackhammer (JA), siren (SI), and street music (ST)



Analysis of Representations, Weights

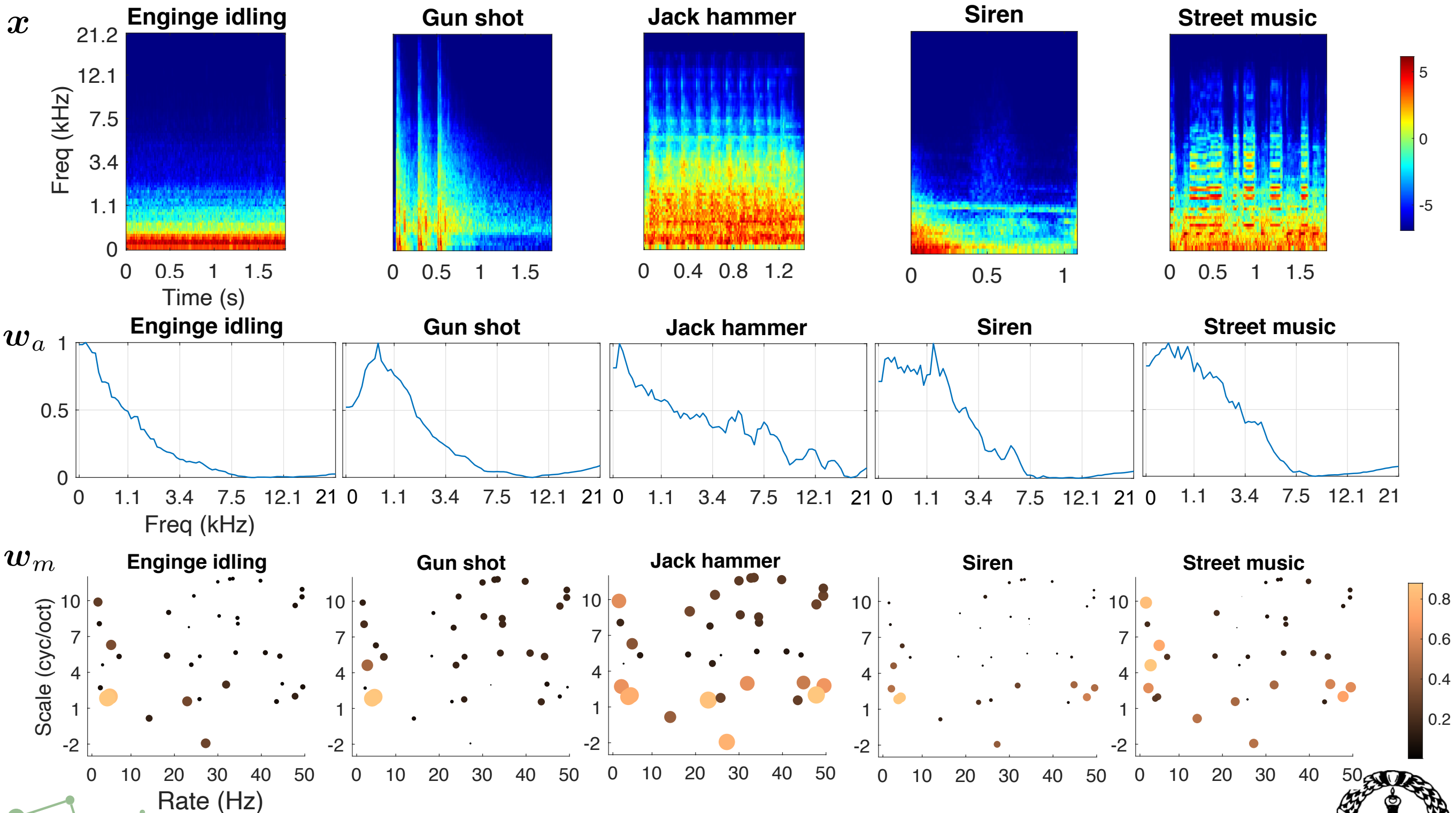
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- Observing x , w_a and w_m



Analysis of Representations, Weights

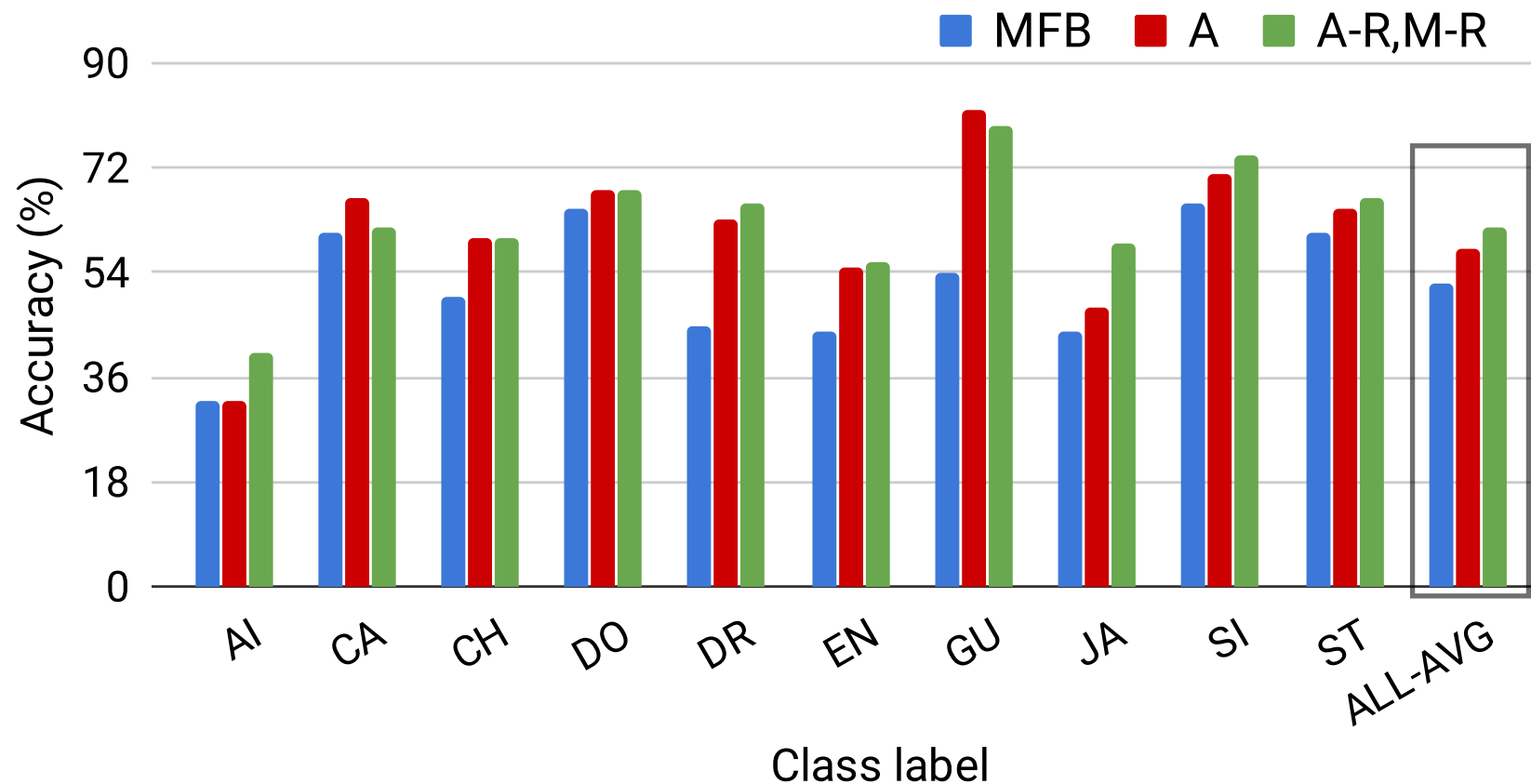
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Experiments With Urban Sounds

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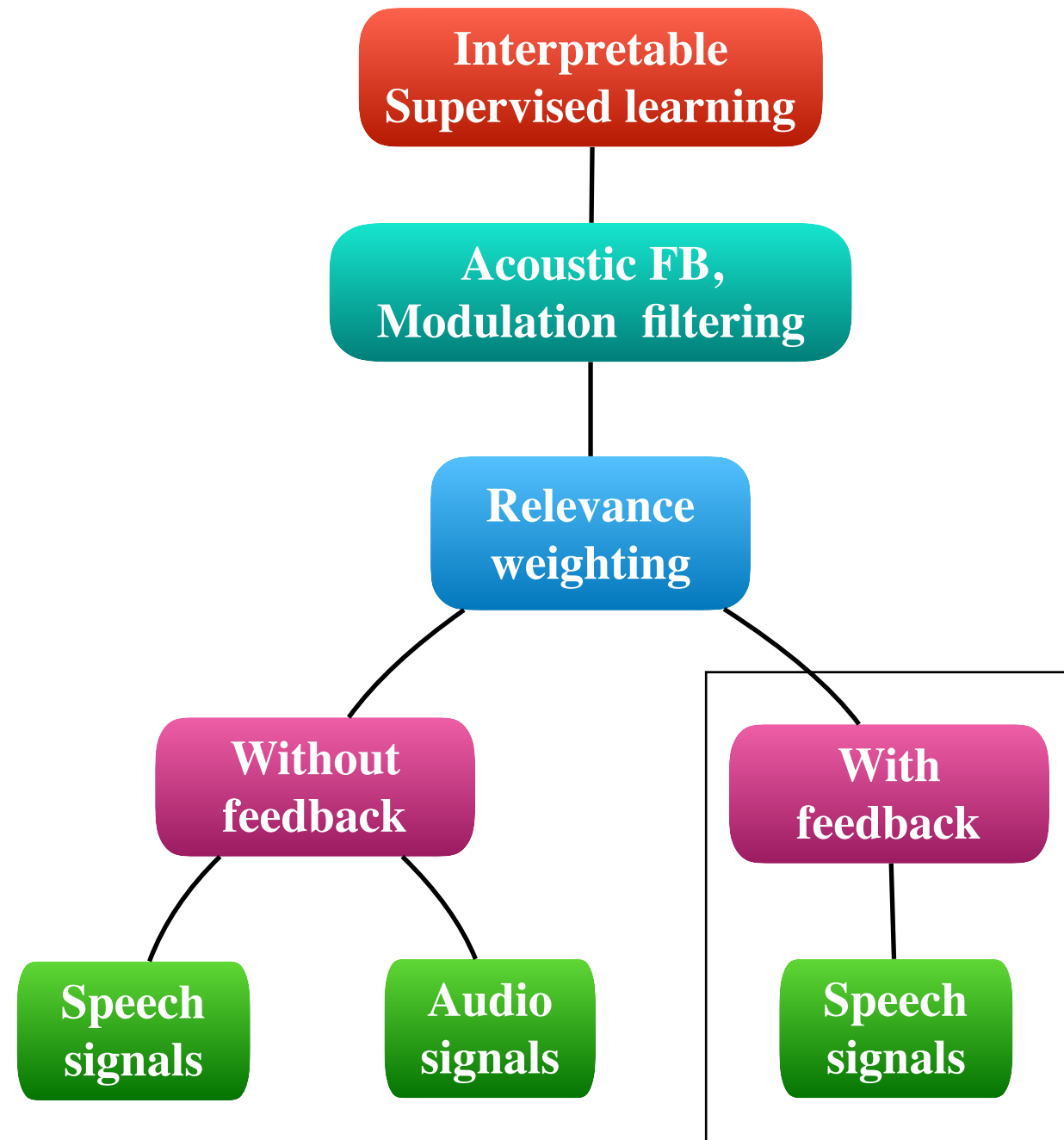
- UrbanSound8K dataset - 8732 labeled real-time sound excerpts (≤ 4 s)
- Urban sounds - air conditioner (AI), car horn (CA), children playing (CH), dog bark (DO), drilling (DR), engine idling (EN), gun shot (GU), jackhammer (JA), siren (SI), and street music (ST)



Absolute improvement of 6% in classification accuracy with learnt acoustic filterbank (A), 10% with proposed A-R,M-R approach

Part II: Supervised Learning (cont...)

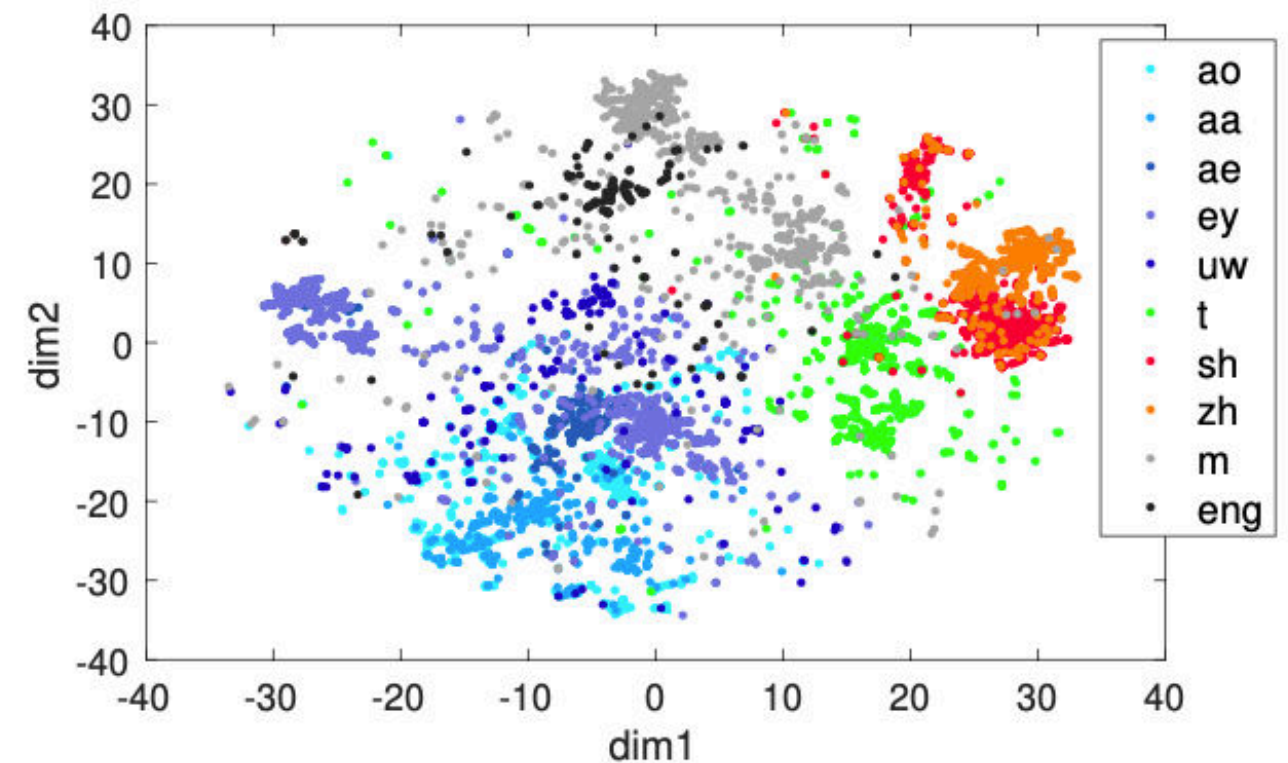
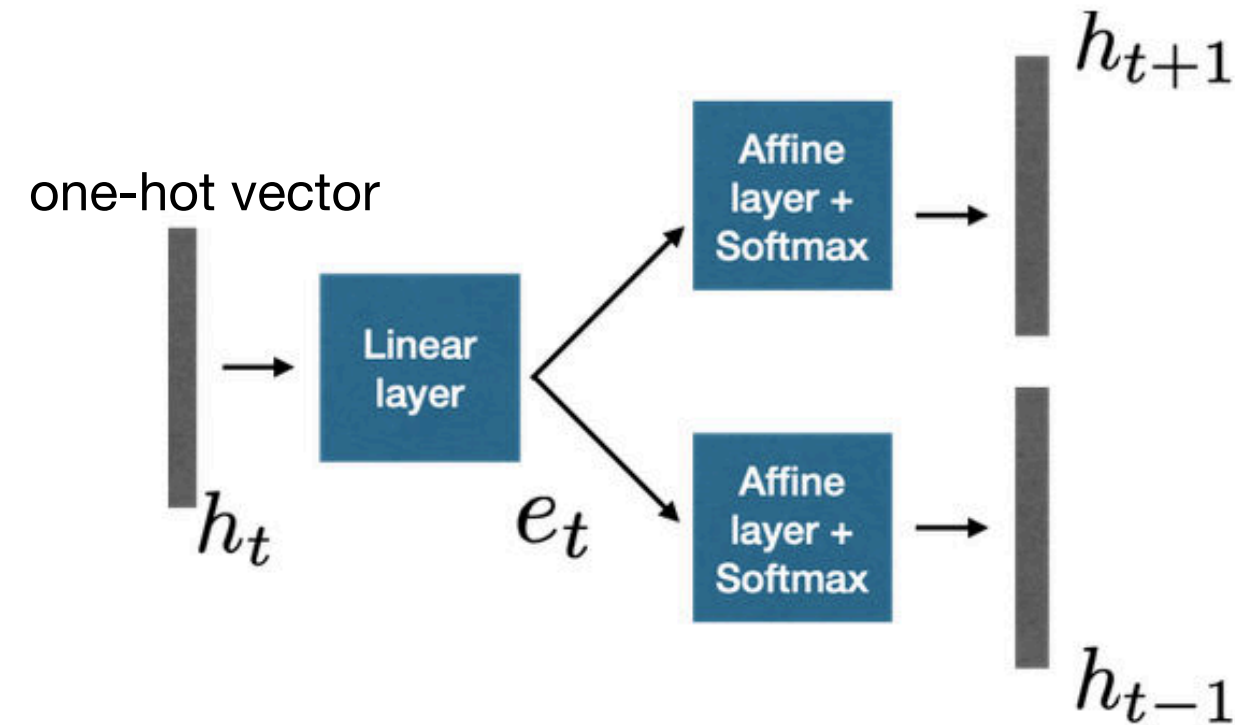
55



Representation Learning With Feedback

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- Feedback of senone embeddings to learn relevance weights



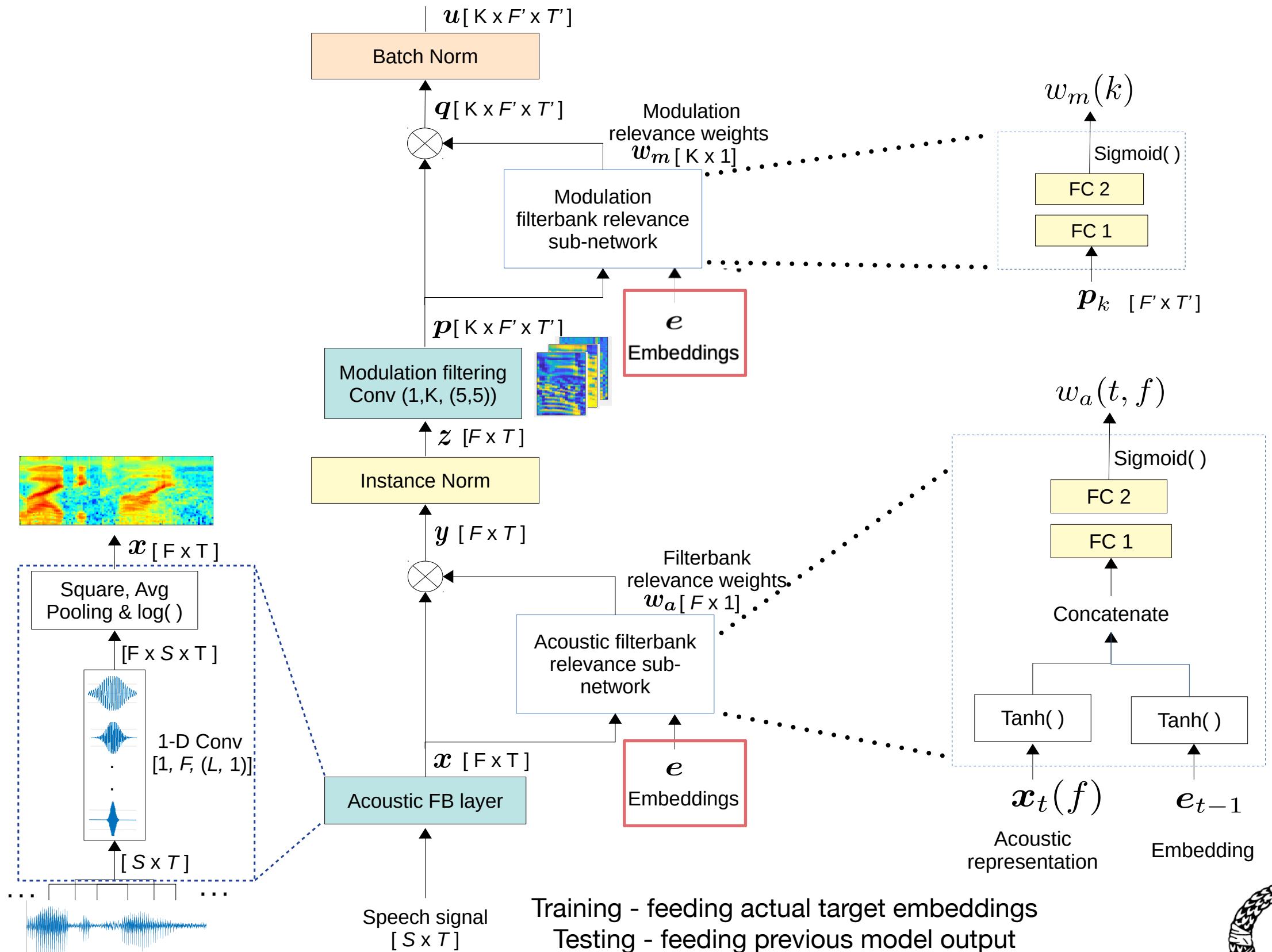
Block schematic of senone embedding network-
Word2vec style

t-SNE plot of the embeddings
for TIMIT dataset
(for visualisation purpose)

- Use teacher forcing network approach
 - Training - Feed actual target embeddings of previous time frame
 - Testing - Feed model outputs back to the network

Representation Learning With Feedback

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Impact on ASR

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**Effect of different components of the discussed model
for the ASR task on Aurora-4 dataset**

Features	ASR (WER in %)
A (Baseline with learnt acoustic FB)	10.7
Acoustic Relevance	
A-R [Softmax, no embedding]	10.0
A-R [Sigmoid, no embedding]	9.9
A-R [Sigmoid, with senone embedding]	9.6
Acoustic + Mod. Relevance	
A-R,M-R [Softmax, no embedding]	9.6
A-R,M-R [Sigmoid, no embedding]	9.5
A-R, M-R [Sigmoid, with senone embedding]	9.2

Average relative improvement of 14% in WER in multi cond. training

Part II: Summary

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- Part II : Supervised representation learning
 - Proposed 2-stage representation learning framework from raw waveform
 - Parametric approach to learning network layer parameters - 1-D acoustic filterbank, 2-D modulation filterbank
 - Incorporating relevance weighting scheme in both stages - allows the feature representations to be adaptively selected
 - Analysis of the intermediate network representations, weight parameters
 - Extended the 2-stage approach to audio signals for urban sound classification
 - Use of target embeddings as feedback to learn relevance weights

Thesis Summary - Key Takeaways

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- Neural representation learning of the speech data with focus on initial 2 stages
- **Unsupervised learning**
 - Stage-1: Acoustic representation from raw speech waveform
 - Stage-2: Modulation filtered representation from spectrogram features
 - Directly compares unsupervised learning in depth
 - Novelty in learning the irredundant modulation filters from the data
- **Supervised learning - Interpretability**
 - Parametric approach to acoustic and modulation representation
 - Proposing to incorporate relevance weighting scheme to acoustic/modulation filterbank
 - Analyse network parameters at intermediate stages
 - Novelty in feedback based modelling with relevance weight
 - Extending the approach to audio signals - Generic learning framework for audio signals

Limitations & Future Work

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Limitations of the study

- Proposed approaches are empirical
 - A model is proposed and analyzed based on performance metrics of ASR
- Computational time
 - Moderate increase in computational time over the baseline features

Future work: The work carried out opens the door to:

- Explore the representation learning approach for
 - Speaker characterisation
 - Language characterisation
- Extending to non-speech sounds such as cough, cry, laughter, etc.
- Incorporating the representation learning for language modelling (LM)
- Self-supervised approach to representation learning

Publications From This Thesis

62

Journals

- P. Agrawal and S. Ganapathy, "Interpretable acoustic and modulation filter learning for ASR", under review, Trans. of Audio, Speech and Lang. Processing, 2020.
- P. Agrawal and S. Ganapathy, "Modulation filter learning using deep variational networks for robust speech recognition," IEEE Journal of Selected Topics in Signal Processing, Special Issue on Data Science: Machine Learning for Audio Signal Processing, 2019.
- P. Agrawal and S. Ganapathy, "Unsupervised modulation filter learning for noise-robust speech recognition," Journal of Acoustical Society of America, 2017.

Conferences

- P. Agrawal, S. Ganapathy, "Interpretable Representation Learning for Speech and Audio Signal Analysis", under review, INTERSPEECH, 2020.
- P. Agrawal, S. Ganapathy, "Robust Raw Waveform Speech Recognition Using Relevance Weighted Representations", under review, INTERSPEECH, 2020.
- P. Agrawal and S. Ganapathy, "Unsupervised raw waveform representation learning for ASR", INTERSPEECH, 2019.
- P. Agrawal and S. Ganapathy, "Deep variational filter learning models for speech recognition," IEEE International Conf on Acoustics, Speech, and Signal Processing, ICASSP, 2019.
- P. Agrawal and S. Ganapathy, "Comparison of unsupervised modulation filter learning methods for ASR," INTERSPEECH, 2018.
- N. Takahashi, P. Agrawal, N. Goswami and Y. Mitsufuji, "PhaseNet: Discretized phase modeling with deep neural networks for audio source separation," INTERSPEECH, 2018.
- P. Agrawal and S. Ganapathy, "Speech representation learning using unsupervised data-driven modulation filtering for robust ASR," INTERSPEECH, 2017.

Challenges

- "Low Resource Speech Recognition Challenge for Indian Languages," by Microsoft Research 2018 - Finished 3rd on leaderboard.
- "The 5th CHiME Speech Separation and Recognition Challenge 2018" - Finished in top 10 teams.



https://github.com/iiscleap/CVAE_FilterLearning

https://github.com/iiscleap/Unsupervised_modFilt_CRBM



Thanks

For any queries: reach me at purvia@iisc.ac.in

Webpage: <http://leap.ee.iisc.ac.in/purvi/>

“By far, the greatest danger of Artificial Intelligence is that people conclude too early that they understand it.” – Eliezer Yudkowsky